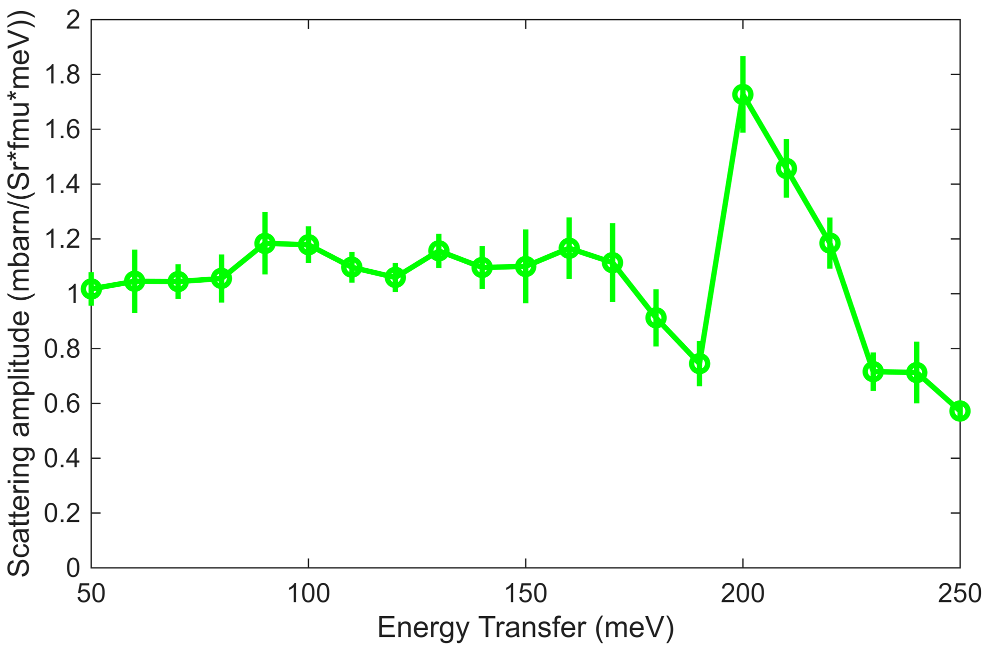
```
fit_data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\Sym4D_cutsAndFits';
% 2 primary fits 1 for 110 and another one for 200 tangential
% directions:
fit_data_file = 'FitEn_cut400ref110_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-
exponent fit fixed bg at high momentum.
fit_data_file = 'FitEn_cut400ref200_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-
exponent fit fixed bg at high momentum.

if ~exist('fit_resEi400','var')
    %fit_data_file = 'FitEn_cut200_1Dsbg_dirGH_dE20_fix_bg_slope.mat'; % fit 1D
background with no Q-slope
    %fit_data_file =
'FitEn_cut200_1D_fit_bg_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-exponent fit
fixed bg at high momentum.

    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi400')
        fit_resEi400 = ld.fit_resEi400;
    else
        fit_resEi400 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_file = 'multicuts_fit_dataGH_ei400meV.mat';
if ~exist('cuts2fit400','var')
    ld = load(fullfile(data_path,fit_file));
    cuts2fit400 = ld.cuts2fit400;
end

mi = maps_instrument(400,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
cl = cuts2fit400.cutsF_list;
for i=1:numel(cl)
    cl{i} = cl{i}.set_instrument(mi);
    cl{i} = cl{i}.set_sample(sample);

    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end
cuts2fit400.cut110_sel = cl{1};
cuts2fit400.cut200_sel = cl{2}
```

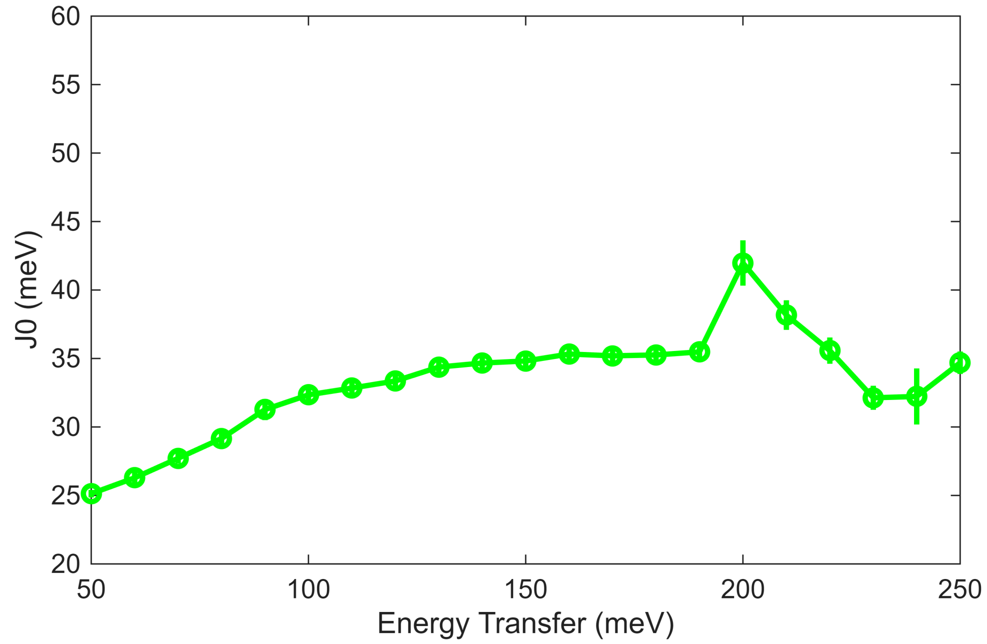
```
cuts2fit400 = struct with fields:
           cuts_list: {[1×1 sqw]  [1×1 sqw]  [1×1 sqw]}
            mask_ds2: {[2×13 double]  [2×9 double]}
            mask_ds3: {[2×9 double]  [2×5 double]}
            mask_ds1: {[2×13 double]  [2×7 double]}
    bg_par_010off100: [1×1 struct]
    bg_par_010off200: [1×1 struct]
    bg_par_010off000: [1×1 struct]
    msk_of100_dir010: {[8×2 double]  [18×2 double]}
          cutsF_list: {[1×1 sqw]  [1×1 sqw]  [1×1 sqw]}
          cut110_sel: [1×1 sqw]
          cutsS_list: {[1×1 sqw]  [1×1 sqw]  [1×1 sqw]}
    msk_of200_dir010: {[12×2 double]  [9×2 double]  [8×2 double]}
```

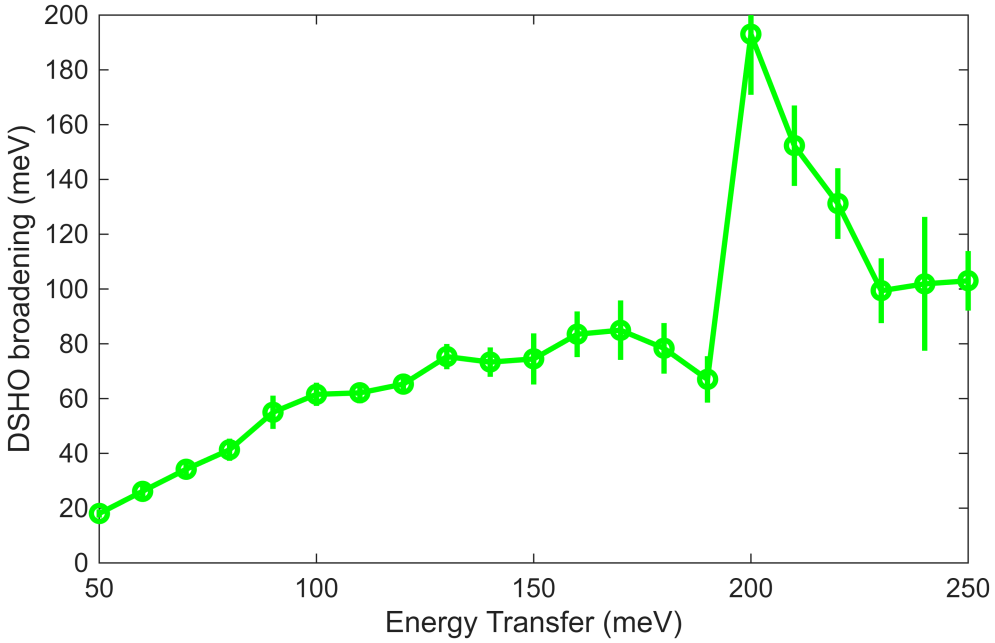
```
msk_of020_dir100: {[12×2 double]  [10×2 double]  [6×2 double]}
      cut200_sel: [1×1 sqw]
```

```
cuts2fit = cuts2fit400;
fit_res = fit_resEi400;
```

```
[Sgnl,J0,G] = plot_SJG_results(fit_res,'g',{'S','J0','gamma'},{[0,2],[20,60],
[0,200]});
```








```
dE = min(Sgnl.x(2:end)-Sgnl.x(1:end-1))
```

```
dE =
10
```

```
e_min = Sgnl.x(1);
```

```
En = 200
```

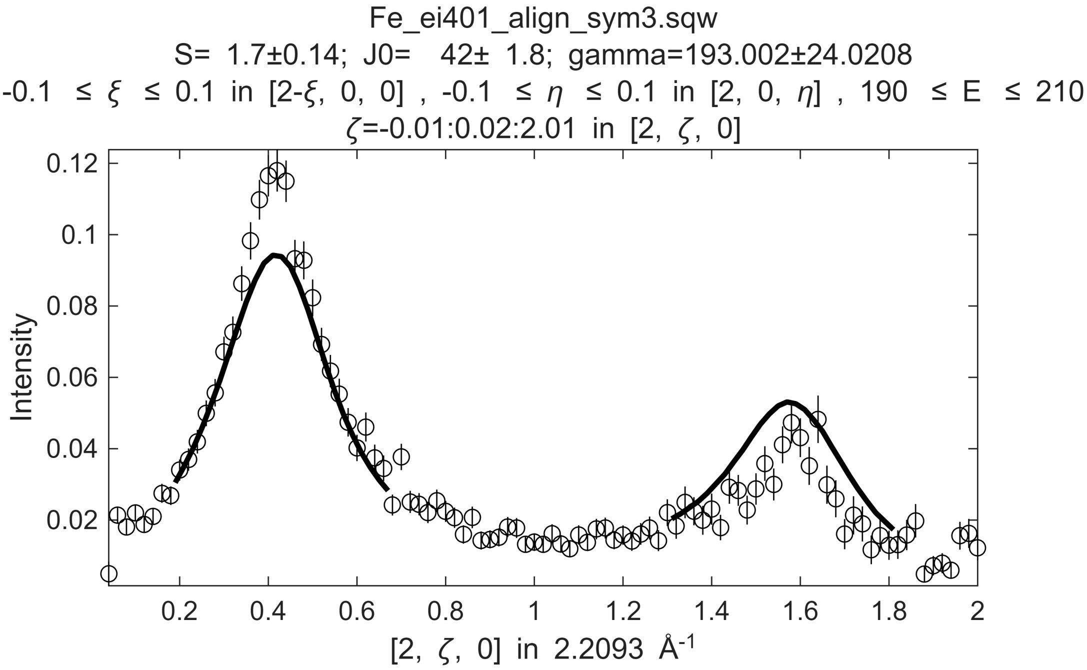
```
En =
200
```

```
fit_par = fit_res.all_fit_par(1);
dE_step = fit_par.en_range(2); %original energy transfer step data were binned
to. No point in going finer
half_dE = 0.5*(fit_par.en_range(3)-fit_par.en_range(1));
en_idx =  floor((En-e_min)/dE)+1;
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);
%the2Dcuts = {cut(cuts2fit.cut110_sel,[0,0.02,2],[])};%(1:2);
the2Dcuts = {cuts2fit.cut200_sel};

%init_bg_params =
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg
%in case of 2-D fitting
%
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p
arams,init_bg_params,false)
lfit_par = fit_res.all_fit_par(en_idx);
en_range = lfit_par.en_range;
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.2
,true,false);
```

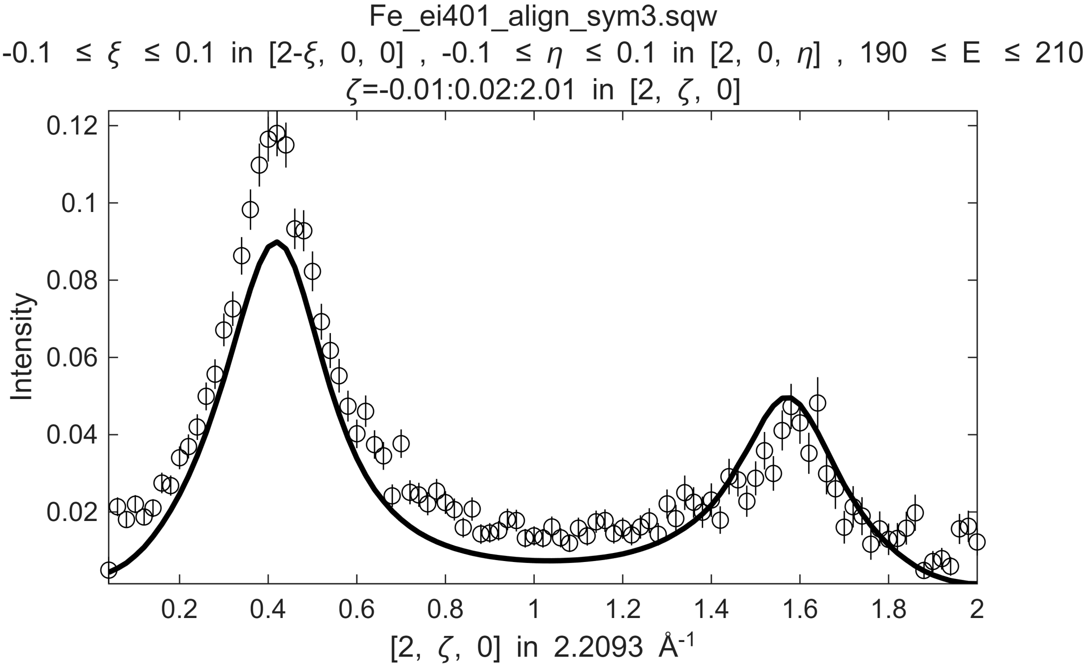
```
Warning: Dataset:12 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.9802
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   542922 pixels -- processing data... -----> included    448914 pixels
--------------------------------------------------------------------------------
Number of points in input object: 786193
     Fraction of object selected:    69.06% (=    542922 points) ;  Access speed:     0.90GB/sec
     Fraction of object retained:    57.10% (=    448914 points)

**** Total time in cut(sqw)    :      0.1sec Including:
     Data access time fraction:  62.6%; Transf. time frac.:  33.8%; Resulting pix preparation:   0.0%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for   316198 pixels -- processing data... -----> retained    304328 pixels
--------------------------------------------------------------------------------
Number of points in input object: 792827
     Fraction of object selected:    39.88% (=    316198 points) ;  Access speed:     1.22GB/sec
     Fraction of object retained:    38.39% (=    304328 points)

**** Total time in cut(sqw)    :      0.1sec Including:
     Data access time fraction:  27.5%; Transf. time frac.:  45.8%; Resulting pix preparation:  26.7%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   259690 pixels -- processing data... -----> included    209029 pixels
--------------------------------------------------------------------------------
Number of points in input object: 786193
     Fraction of object selected:    33.03% (=    259690 points) ;  Access speed:     1.23GB/sec
     Fraction of object retained:    26.59% (=    209029 points)

**** Total time in cut(sqw)    :      0.0sec Including:
     Data access time fraction:  51.7%; Transf. time frac.:  45.6%; Resulting pix preparation:   0.0%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for   115502 pixels -- processing data... -----> retained    111234 pixels
--------------------------------------------------------------------------------
Number of points in input object: 792827
     Fraction of object selected:    14.57% (=    115502 points) ;  Access speed:     0.54GB/sec
     Fraction of object retained:    14.03% (=    111234 points)

**** Total time in cut(sqw)    :      0.0sec Including:
     Data access time fraction:  29.5%; Transf. time frac.:  38.3%; Resulting pix preparation:  32.1%
--------------------------------------------------------------------------------
Warning: Dataset 2:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.9231
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------
Starting point
------------------
 Total time = 6.1333s    Reduced Chi^2 = 2.1619
 Free parameter values:
           193          1.727          41.97       -0.01868       -0.01868
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 28.9191s    Reduced Chi^2 = 3.8809      Levenberg-Marquardt = 0.1
 Free parameter values:
         157.5          1.512          38.94         -3.576          8.979
 Total time = 34.9797s    Reduced Chi^2 = 3.3648      Levenberg-Marquardt = 1
 Free parameter values:
         181.7           1.65          40.92         -2.399          6.723
 Total time = 41.0328s    Reduced Chi^2 = 2.5823      Levenberg-Marquardt = 10
 Free parameter values:
         189.7          1.699          41.68         -1.014          2.856
 Total time = 47.2515s    Reduced Chi^2 = 2.1941      Levenberg-Marquardt = 1000
 Free parameter values:
         192.7          1.724          41.95        -0.1218         0.2832
 Total time = 53.1051s    Reduced Chi^2 = 2.1649      Levenberg-Marquardt = 100000
 Free parameter values:
```

```
          193          1.727          41.97       -0.02899        0.01153
 Total time = 59.1979s    Reduced Chi^2 = 2.1649      Levenberg-Marquardt = 10000000
 Free parameter values:
          193          1.727          41.97       -0.01971       -0.01566
 *** No improvement in chi-squared over previous iteration ***
 Total time = 59.2112s    Reduced Chi^2 = 2.1619
 Free parameter values:
          193          1.727          41.97       -0.01868       -0.01868
--------------------------------------------------------------------------------
Fit converged:

Parameter values (free parameters with error estimates):
--------------------------------------------------------

Foreground parameter values:
----------------------------
    1             1
    2             8
    3           193   +/-   24.02
    4         1.727   +/-   0.136
    5             0
    6         41.97   +/-   1.809
    7             0
    8             0
    9             0
   10             0


Background parameter values:
----------------------------
  Function 1:
    1      -0.01868   +/-   0.732
    2      -0.03409
    3     -0.006085
    4    -0.0001965

  Function 2:
    1      -0.01868   +/-   0.7312
    2      -0.03409
    3     -0.006085
    4    -0.0001965


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9673    0.9566    0.1451   -0.1391
    0.9673    1.0000    0.9457    0.2017   -0.1958
    0.9566    0.9457    1.0000    0.1669   -0.1611
    0.1451    0.2017    0.1669    1.0000   -1.0000
   -0.1391   -0.1958   -0.1611   -1.0000    1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   792827 pixels -- processing data... ----->  included    658312 pixels
--------------------------------------------------------------------------------
Number of points in input object: 792827
     Fraction of object selected:   100.00% (=    792827 points) ;  Access speed:     1.11GB/sec
     Fraction of object retained:    83.03% (=    658312 points)

**** Total time in cut(sqw)   :      0.1sec Including:
     Data access time fraction:  59.6%; Transf. time frac.:  39.4%; Resulting pix preparation:   0.0%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   304328 pixels -- processing data... ----->  included    252745 pixels
--------------------------------------------------------------------------------
Number of points in input object: 304328
     Fraction of object selected:   100.00% (=    304328 points) ;  Access speed:     1.30GB/sec
     Fraction of object retained:    83.05% (=    252745 points)
```

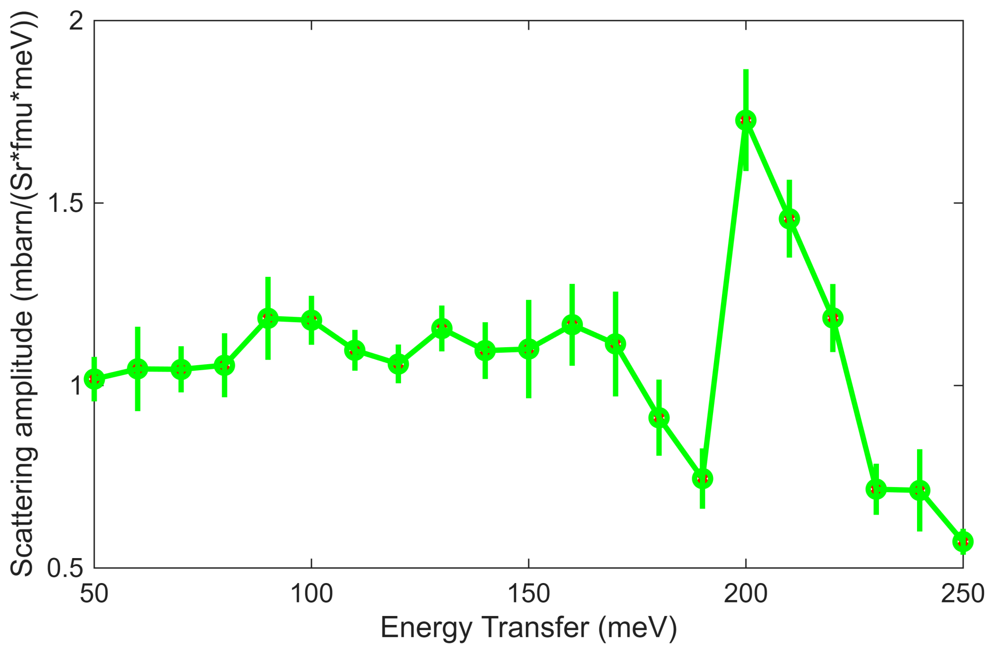
```
**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  46.4%; Transf. time frac.:  51.4%; Resulting pix preparation:   0.0%
-------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for   109914 pixels -- processing data... ----->  included     89988 pixels
-------------------------------------------------------------------------------
Number of points in input object: 109914
     Fraction of object selected:   100.00% (=    109914 points) ;  Access speed:     1.32GB/sec
     Fraction of object retained:    81.87% (=     89988 points)

**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  37.2%; Transf. time frac.:  57.9%; Resulting pix preparation:   0.0%
-------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for   792827 pixels -- processing data... ----->  retained
```

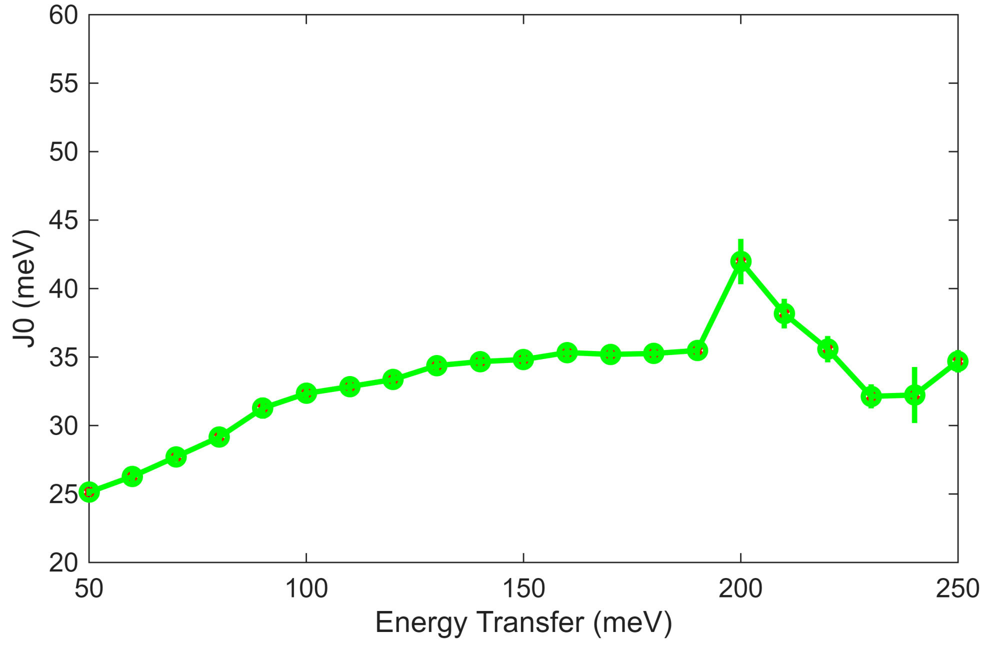


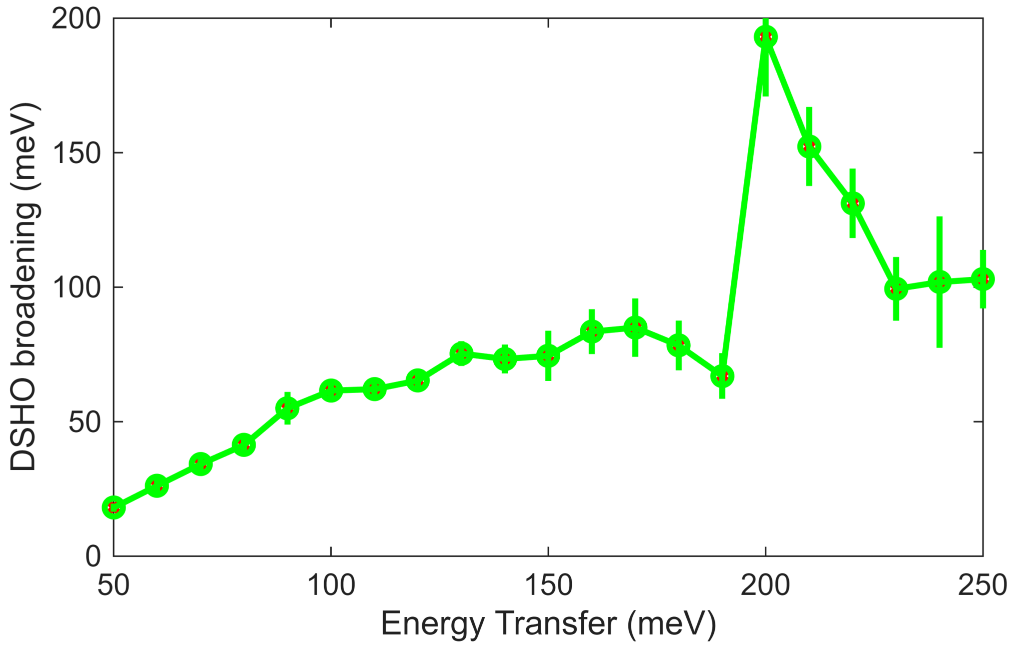

```
    658312 pixels
-------------------------------------------------------------------------------
Number of points in input object: 792827
     Fraction of object selected:   100.00% (=    792827 points) ;  Access speed:     1.26GB/sec
     Fraction of object retained:    83.03% (=    658312 points)

**** Total time in cut(sqw)   :      0.2sec Including:
     Data access time fraction:  26.8%; Transf. time frac.:  53.0%; Resulting pix preparation:  20.1%
-------------------------------------------------------------------------------
```




```
%fit_res.all_fit_par(en_idx) = fit_par;
%fit_res.all_fit_par(en_idx).p = fit_par.p;
Sgnl.signal(en_idx) = fit_par.p(4);
J0.signal(en_idx) = fit_par.p(6);
G.signal(en_idx) = fit_par.p(3);
plot_SJG_results(fit_res,'g',{'S','J0','gamma'},{[0.5,2],[20,60],[0,200]},
{Sgnl,J0,G});
```

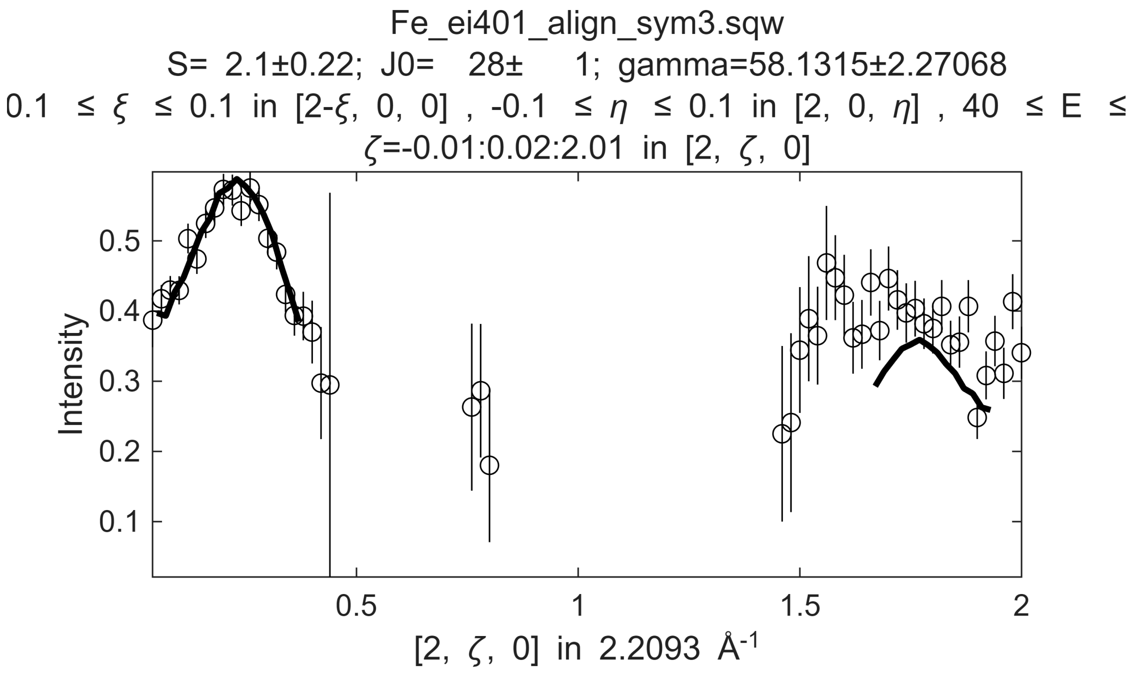


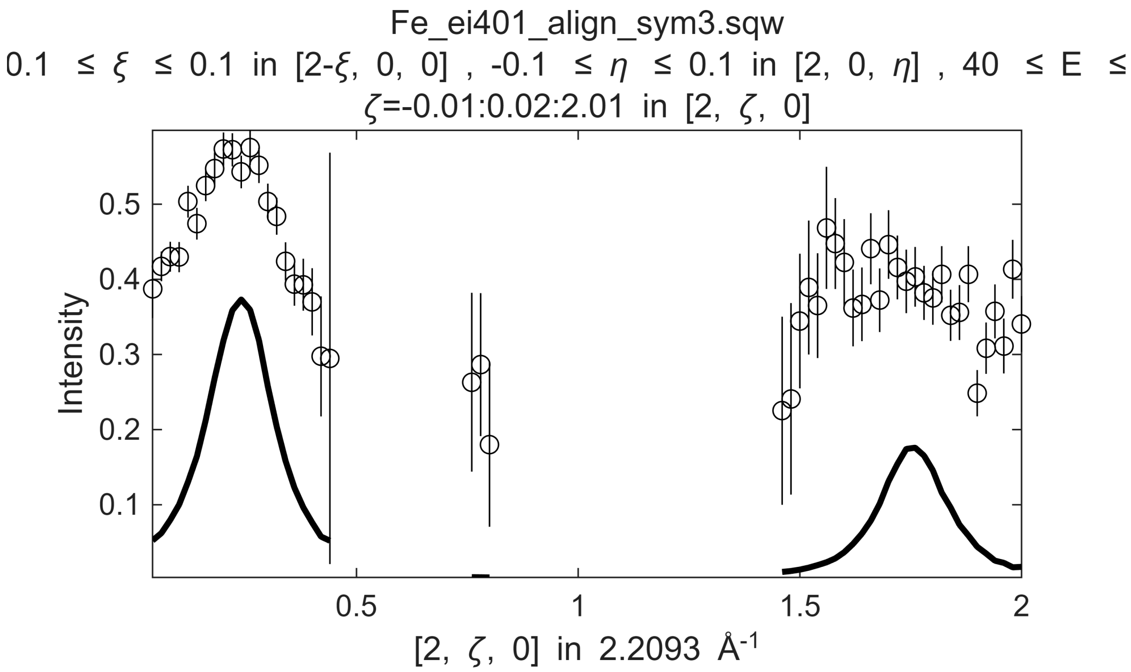





```
[fit_res,Sg,J0,G] = refit_data_around_peaks(the2Dcuts,fit_res,0.2,true);
```

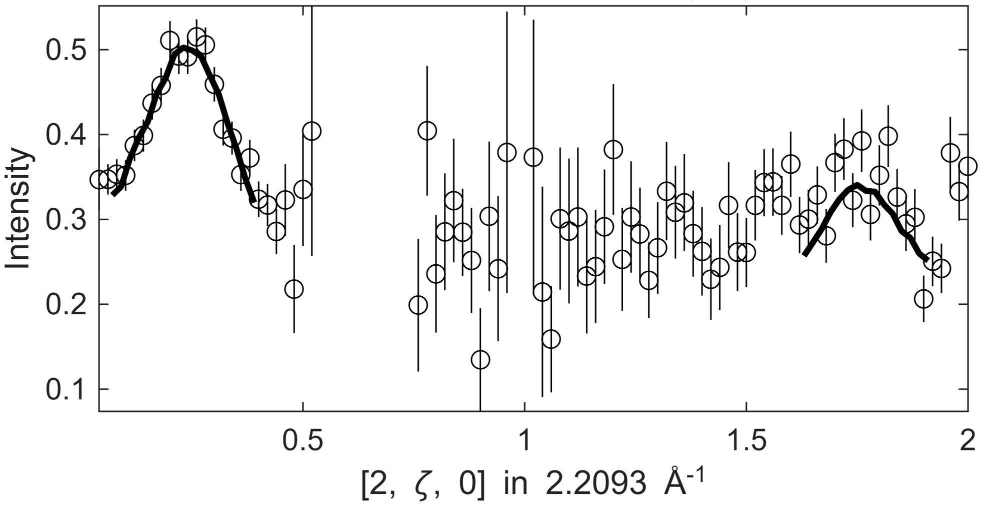
```
***************************************************************
*** step: 1#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
***************************************************************
```



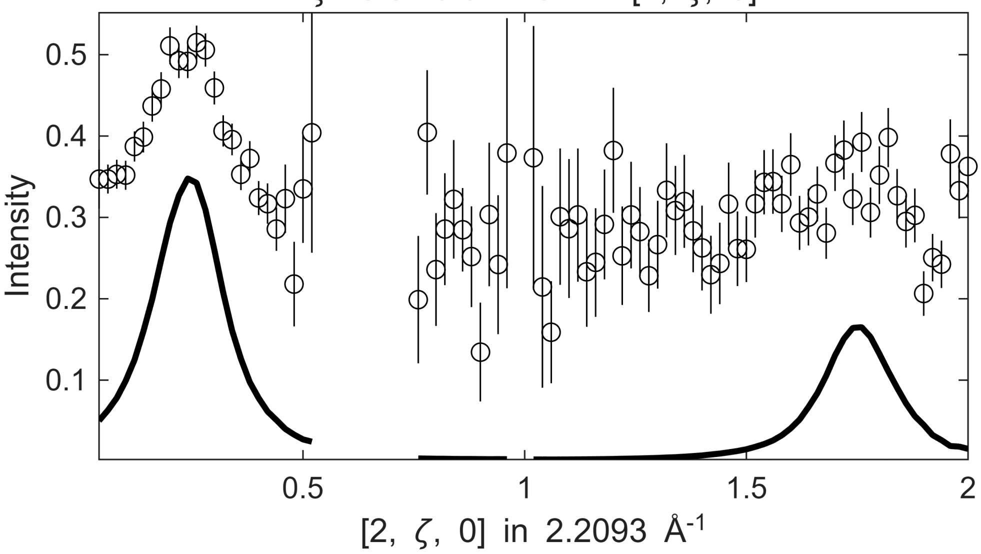



```
*************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*************************************************************
*************************************************************
*** step: 2#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*************************************************************
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.33003
```

Fe_ei401_align_sym3.sqw
S= 2.8± 1.6; J0= 36± 11; gamma=91.7291±72.1979
0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 50 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

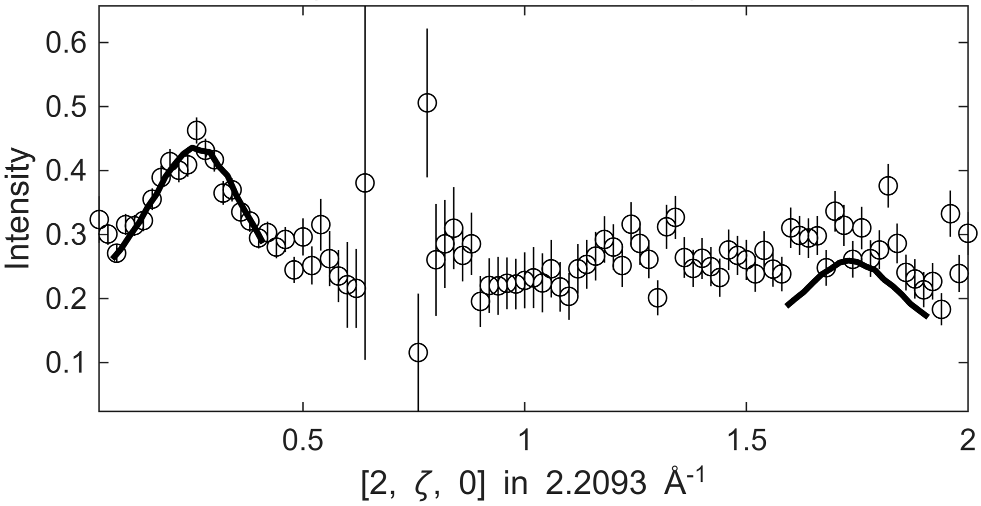


Fe_ei401_align_sym3.sqw
0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 50 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

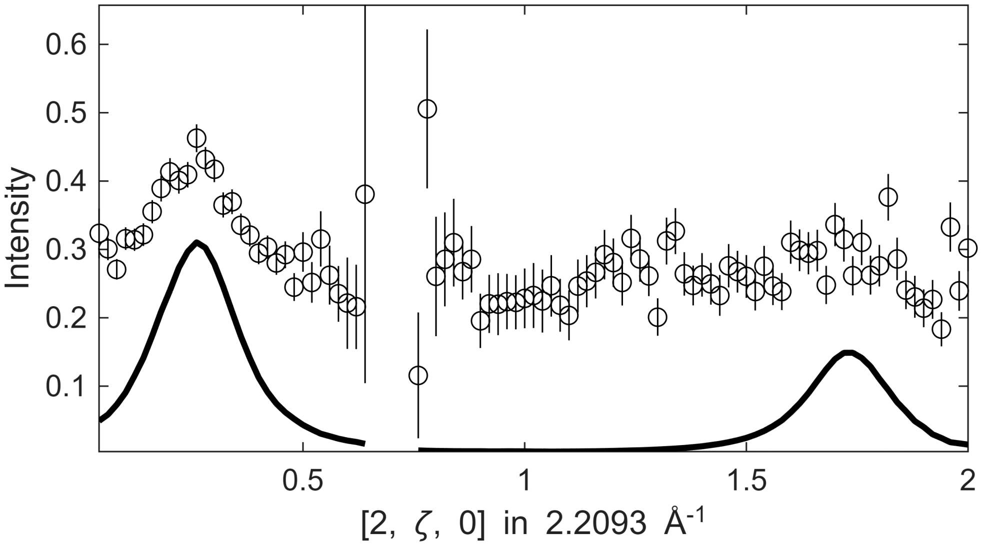



```
*************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*************************************************************
*************************************************************
*** step: 3#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*************************************************************
```

Fe_ei401_align_sym3.sqw
S= 5.4±0.56; J0= 56± 1.8; gamma=281.567±14.8352
0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 60 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 60 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



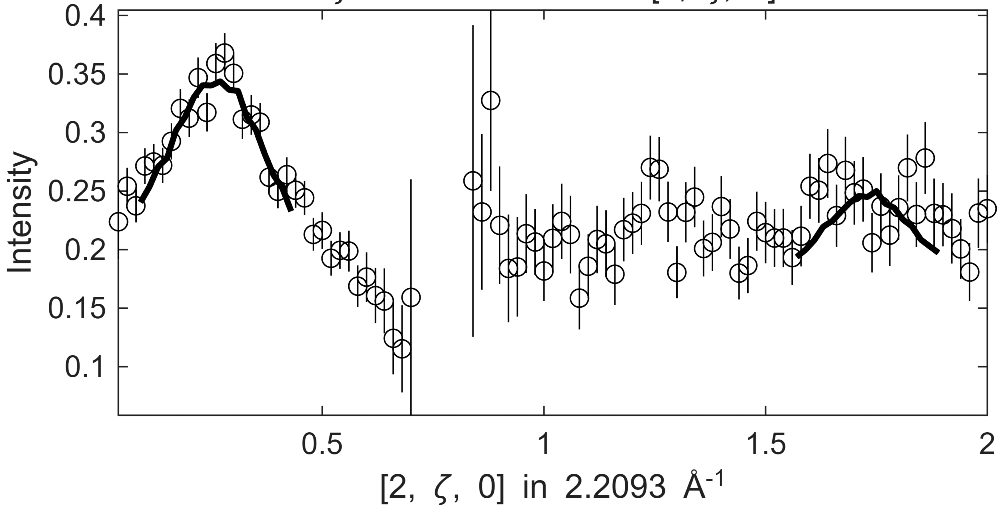

```
**************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
**************************************************************
**************************************************************
*** step: 4#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
**************************************************************
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.33003
```

Fe_ei401_align_sym3.sqw

S= 4.5± 8.5; J0= 68±1e+02; gamma=374.918±1178.53

0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 70 ≤ E ≤

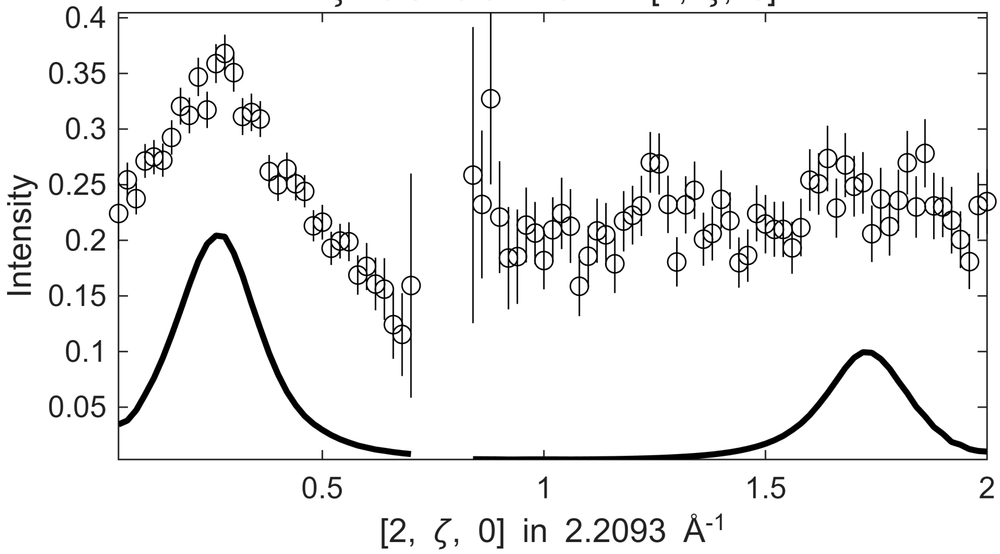
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



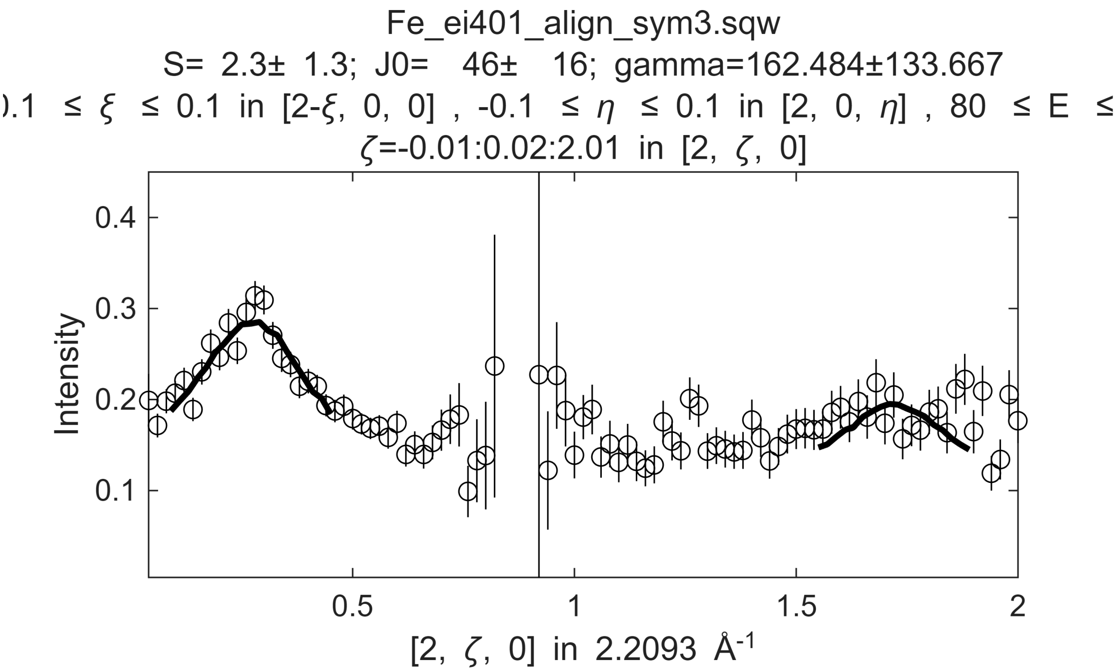
Fe_ei401_align_sym3.sqw

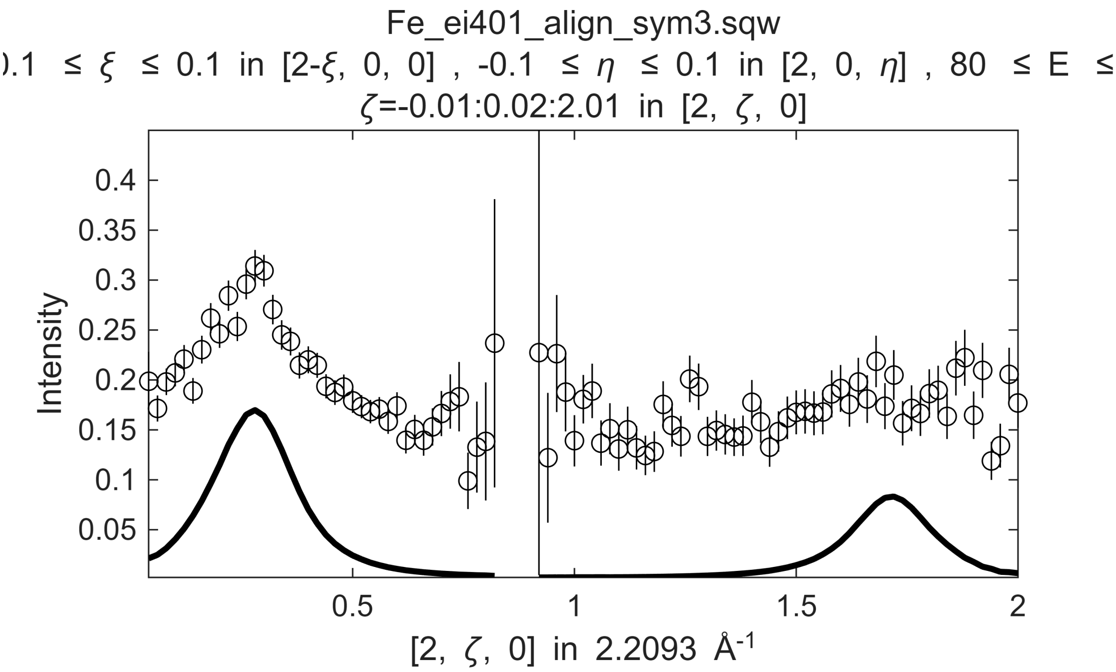
0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 70 ≤ E ≤

ζ=-0.01:0.02:2.01 in [2, ζ, 0]

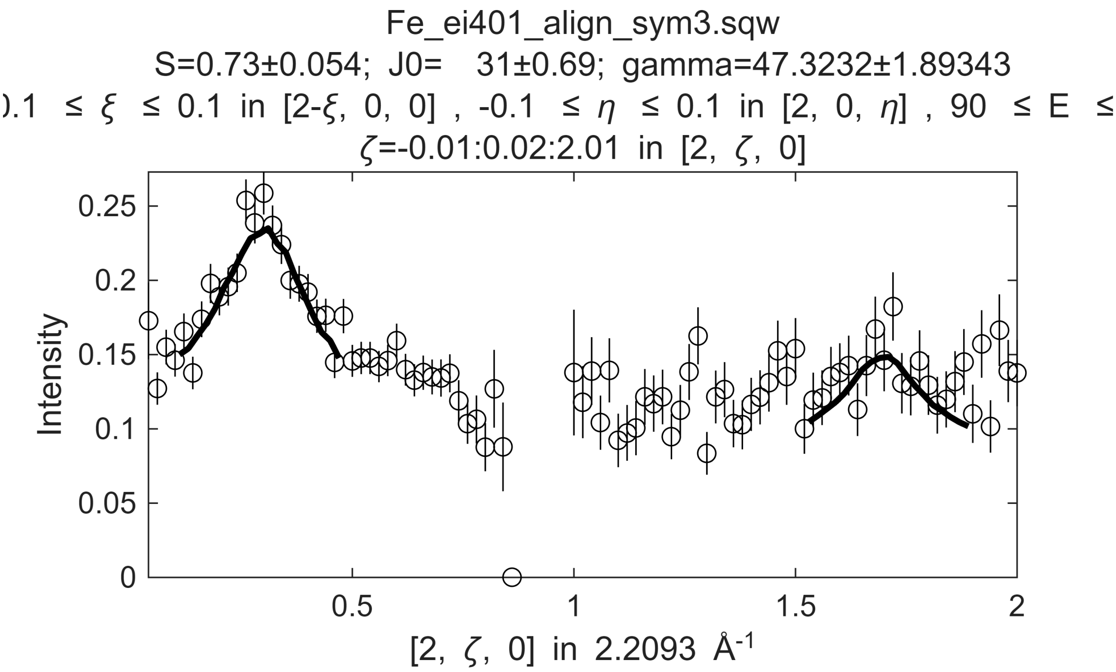


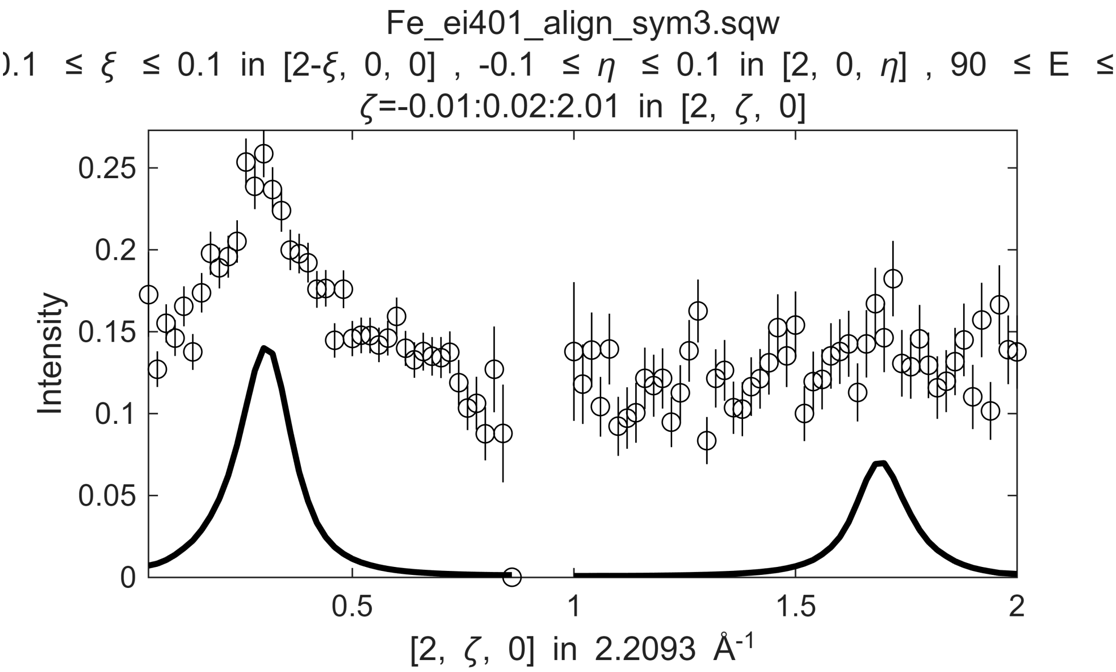

```
*****************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****************************************************************
*****************************************************************
*** step: 5#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****************************************************************
```



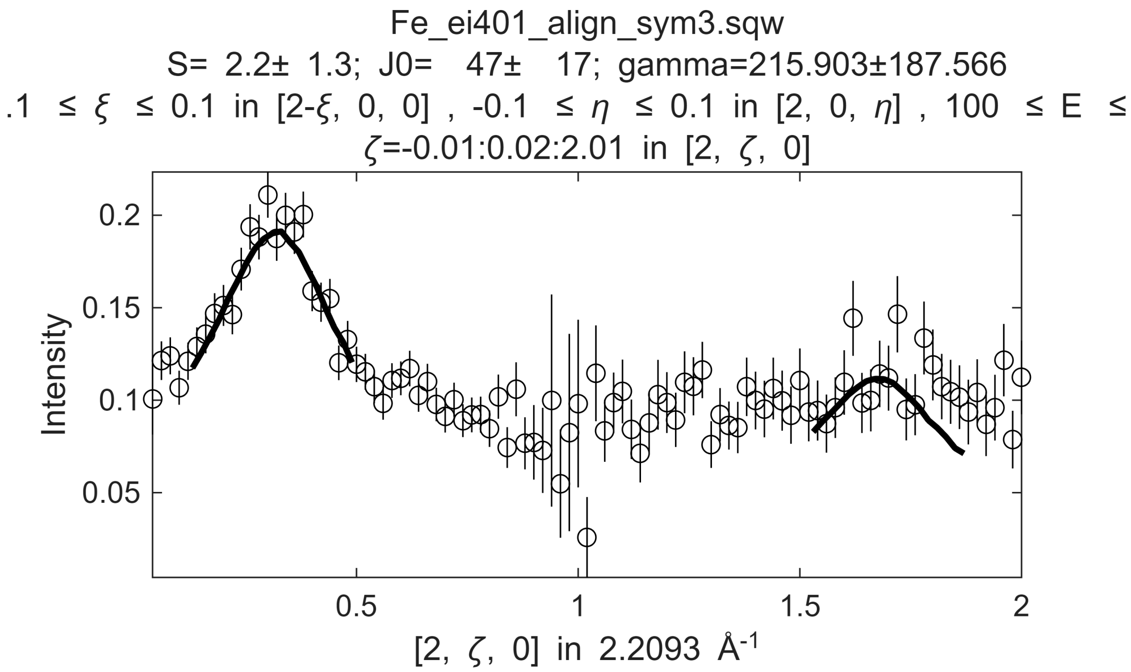


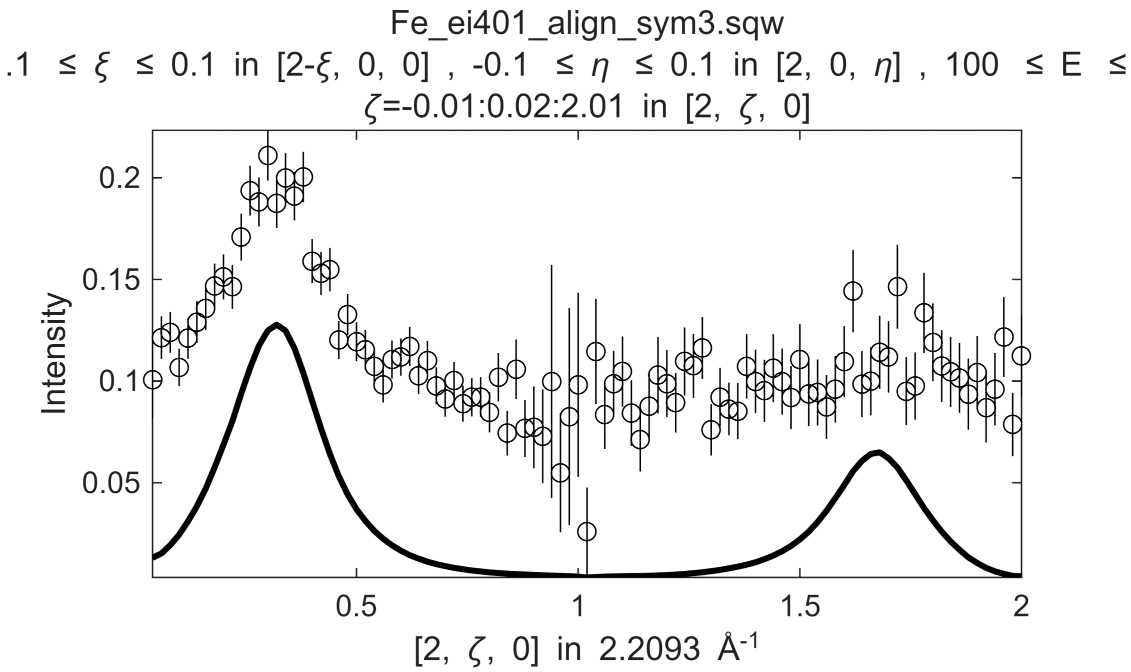

```
************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
************************************************************
************************************************************
*** step: 6#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
************************************************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.16502
```





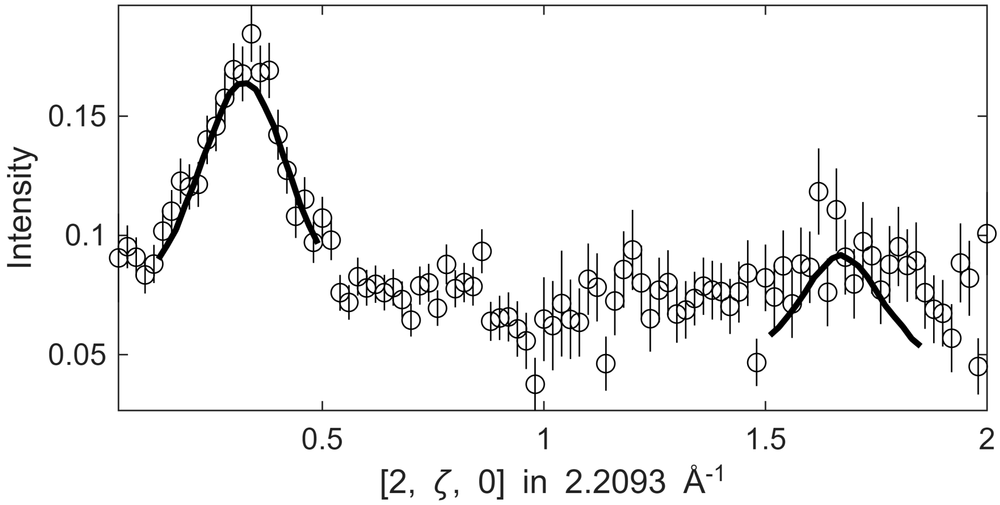

```
*************************************************************
*** &lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt; ***
*************************************************************
*************************************************************
*** step: 7#21  &gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt; ***
*************************************************************
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.33003
```



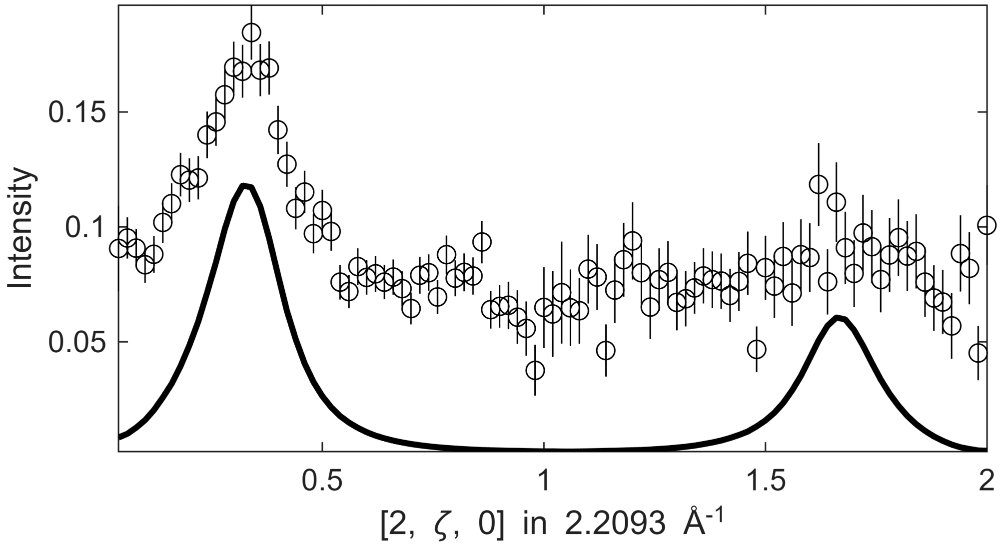



```
************************************************************
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
************************************************************
************************************************************
*** step: 8#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
************************************************************
Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.49505
```

Fe_ei401_align_sym3.sqw
S= 1.2±0.15; J0= 36± 1; gamma=100.35±8.76727
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 110 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 110 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

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*** step: 9#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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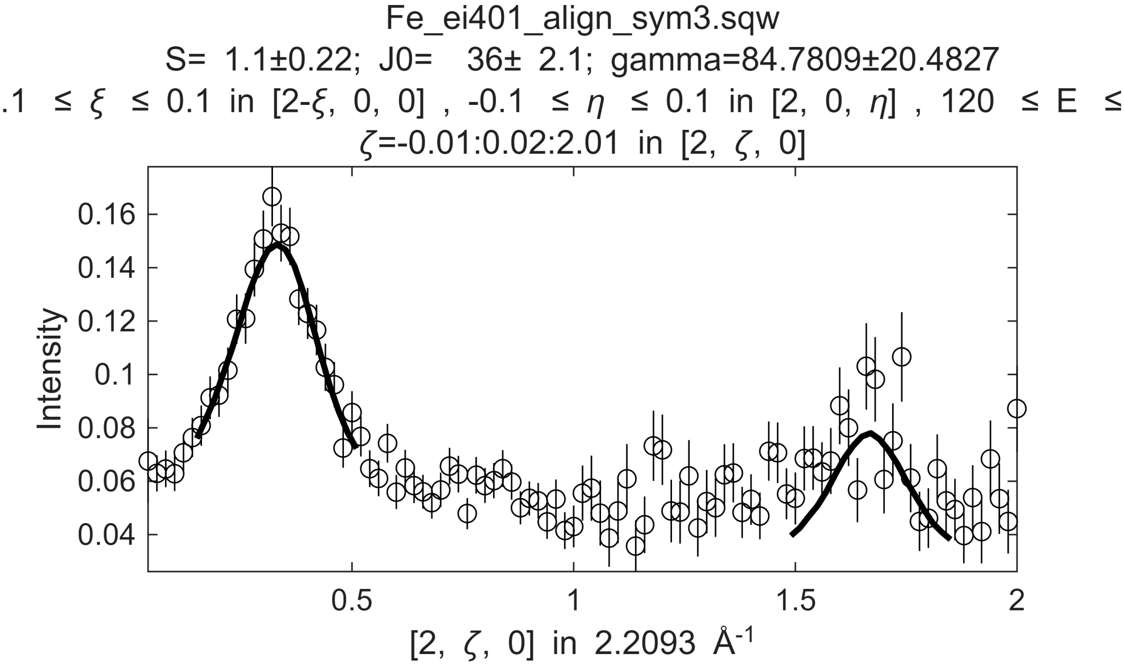
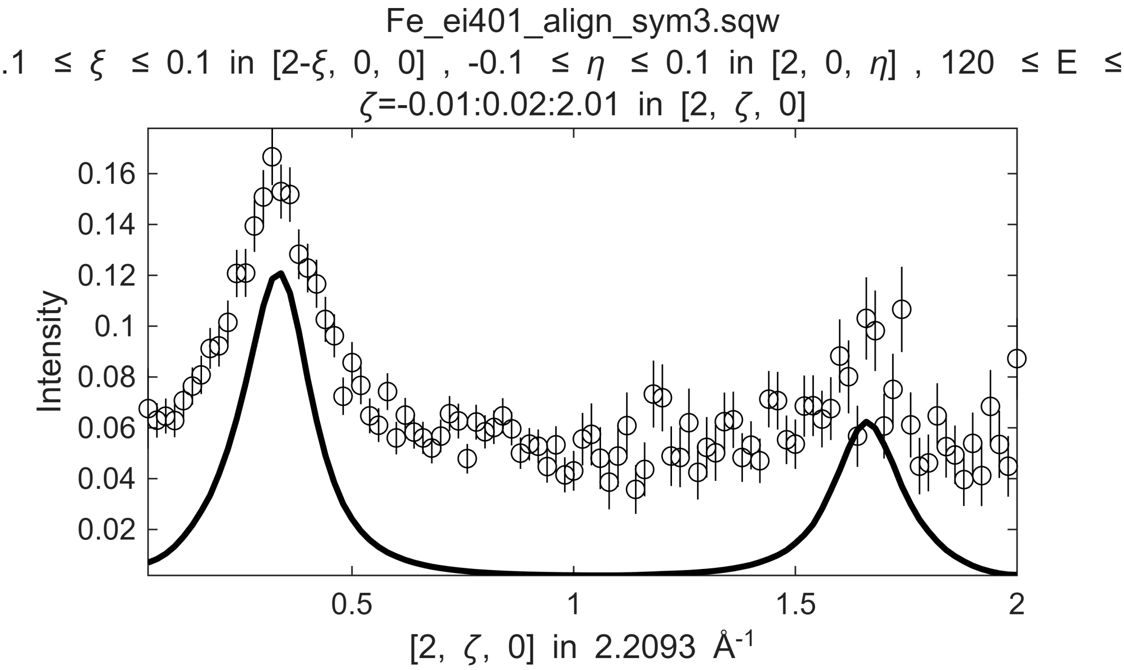
Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.66007
```



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*** step: 10#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:8 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.3201
Warning: Dataset 2:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.83333
```

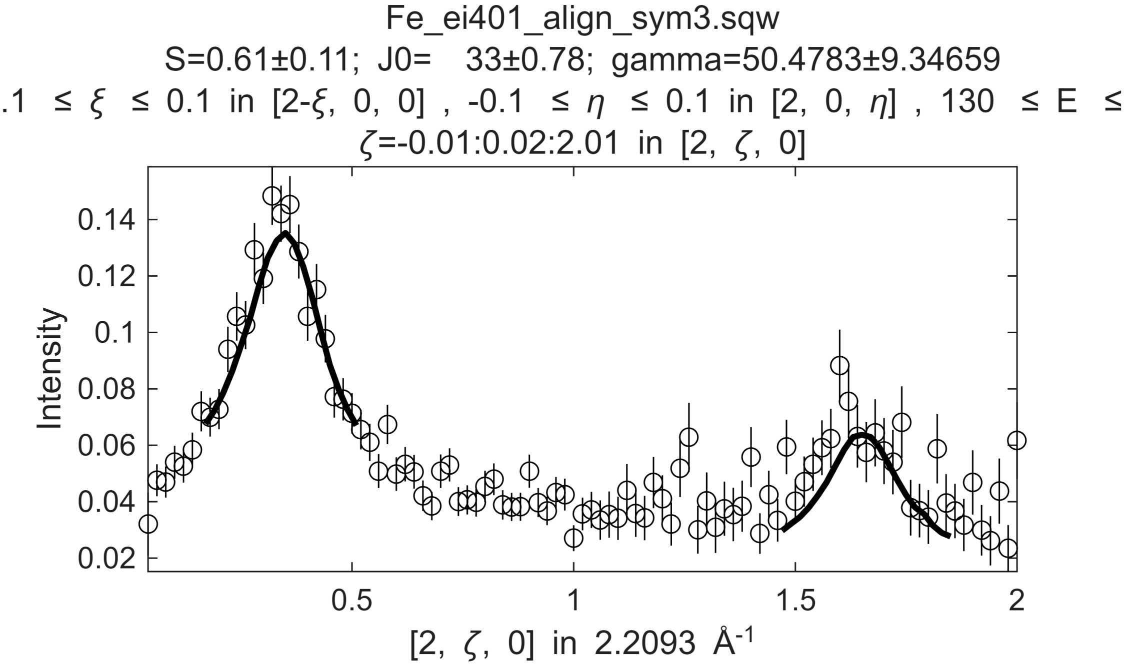

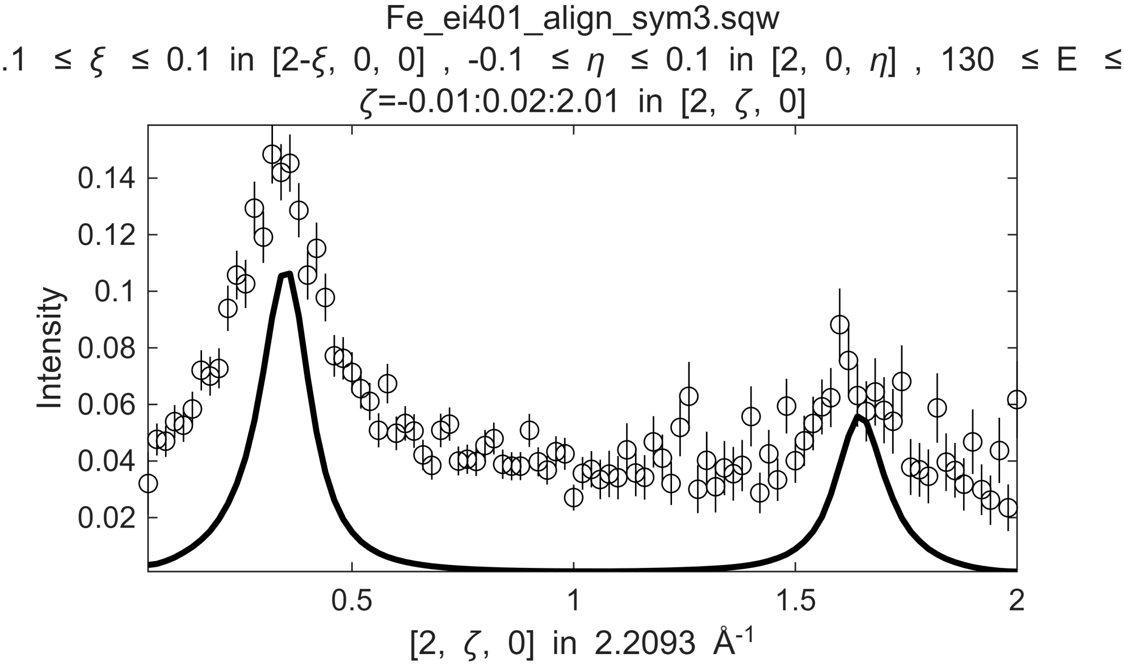


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*** step: 11#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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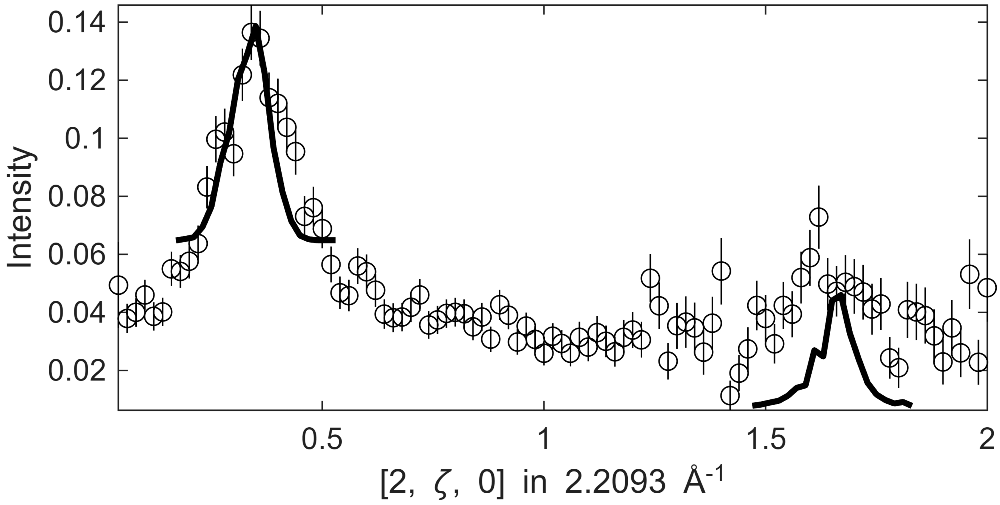
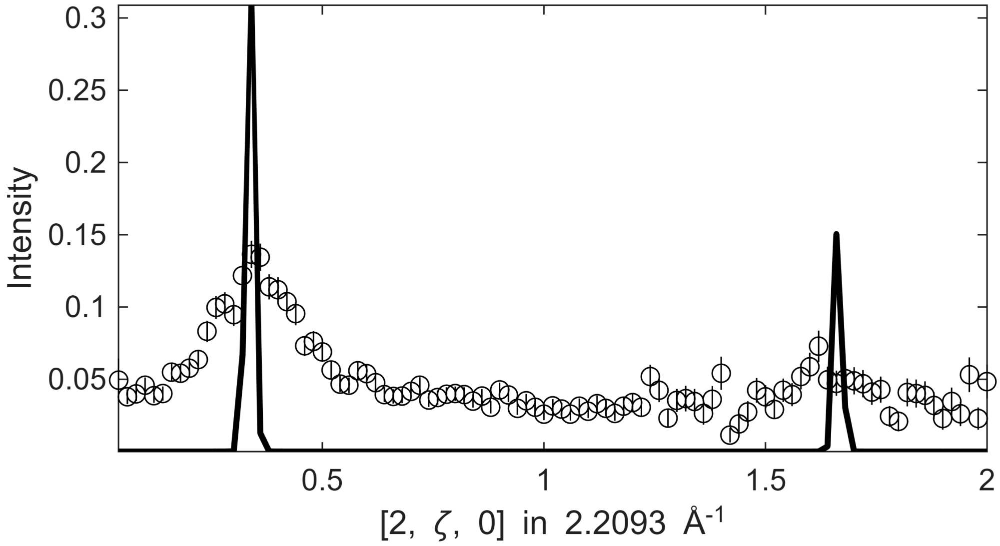
Warning: Dataset:11 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.8152
```

Fe_ei401_align_sym3.sqw
S= 0.2±0.021; J0= 36±0.006; gamma=-0.0470621±0.0267478
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 140 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 140 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

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*** step: 12#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.1551
```

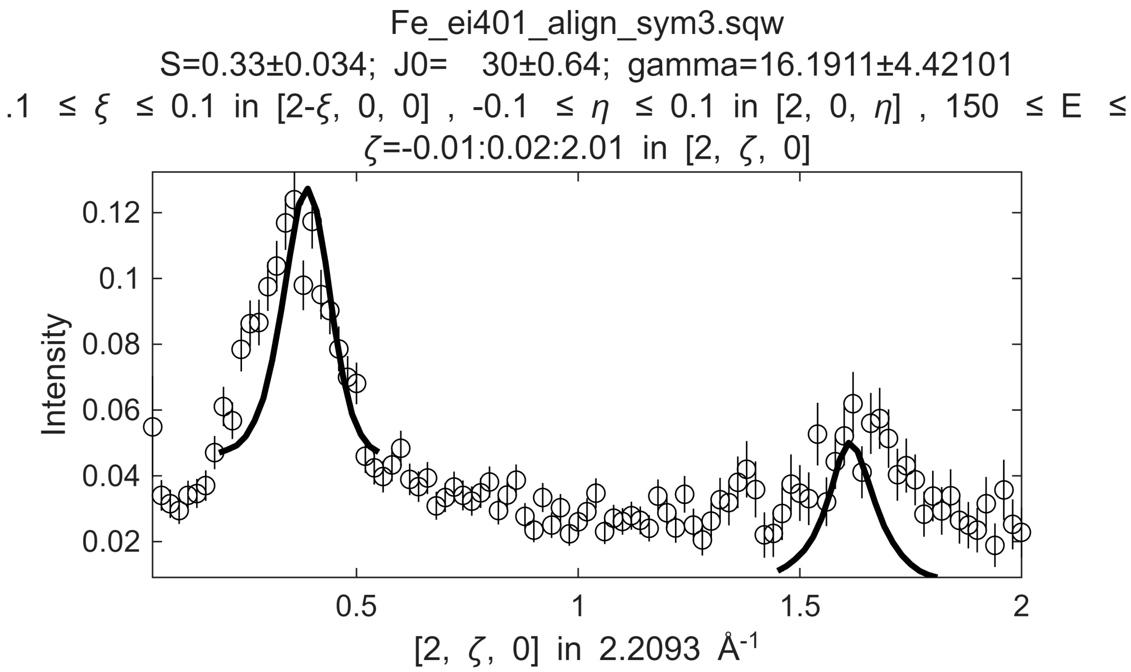

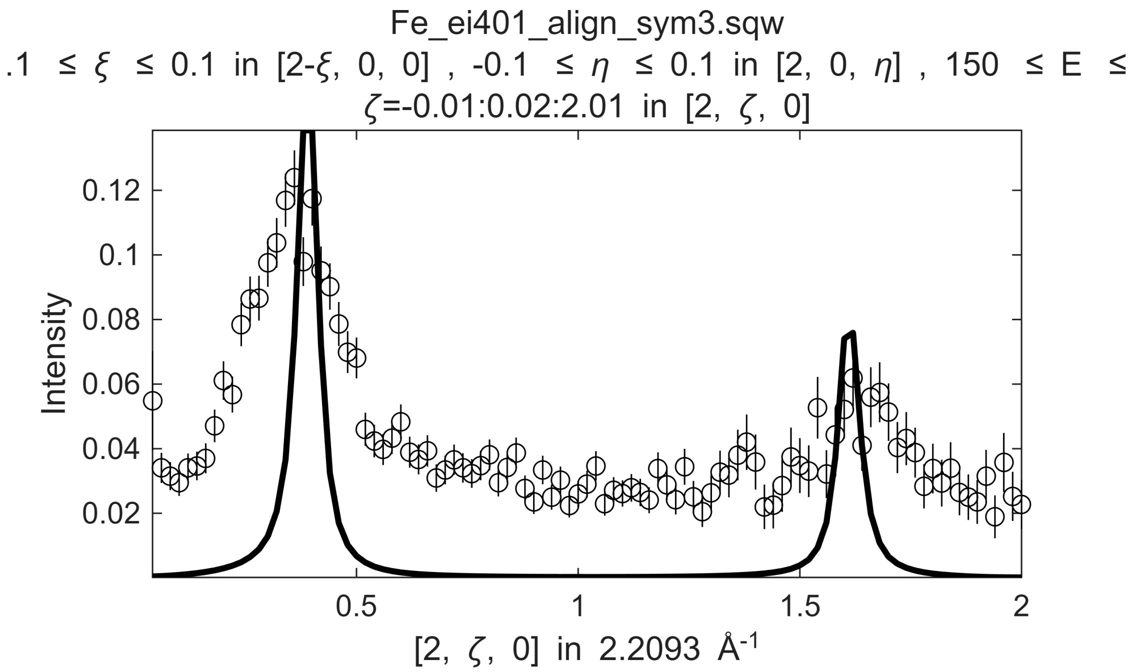


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*** step: 13#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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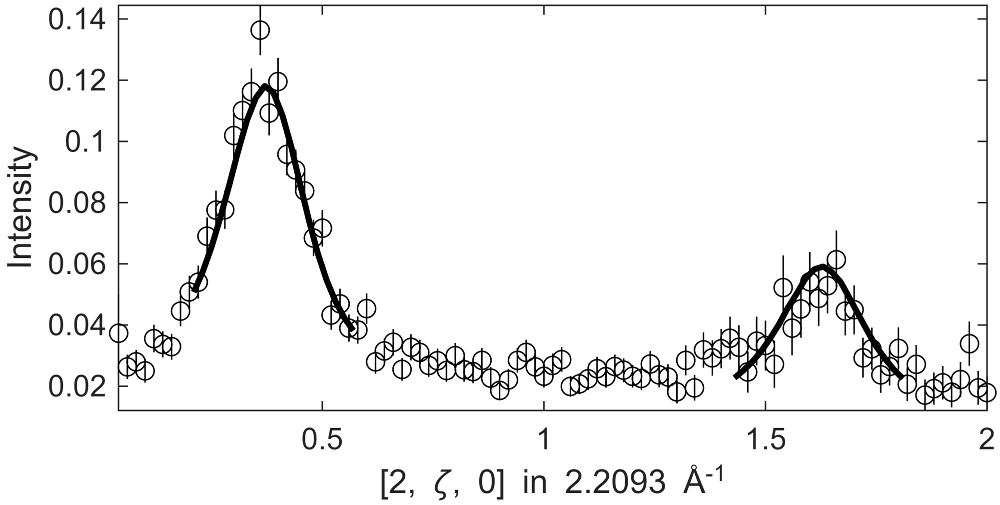
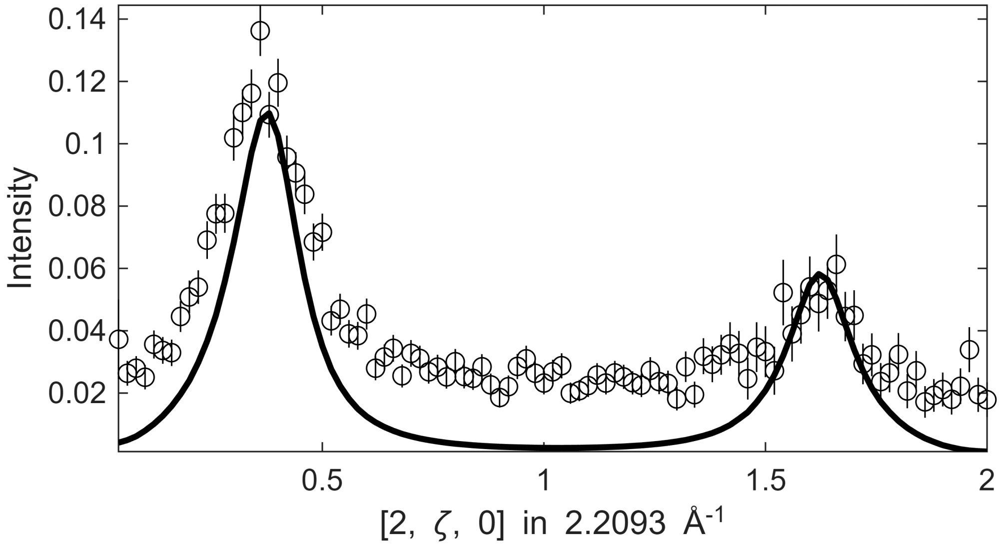
Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.1551
```

Fe_ei401_align_sym3.sqw
S=   1±0.15; J0=   37± 1.1; gamma=85.8472±13.1937
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 160 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 160 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

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*** step: 14#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.1551
Warning: Dataset 2:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.87719
```

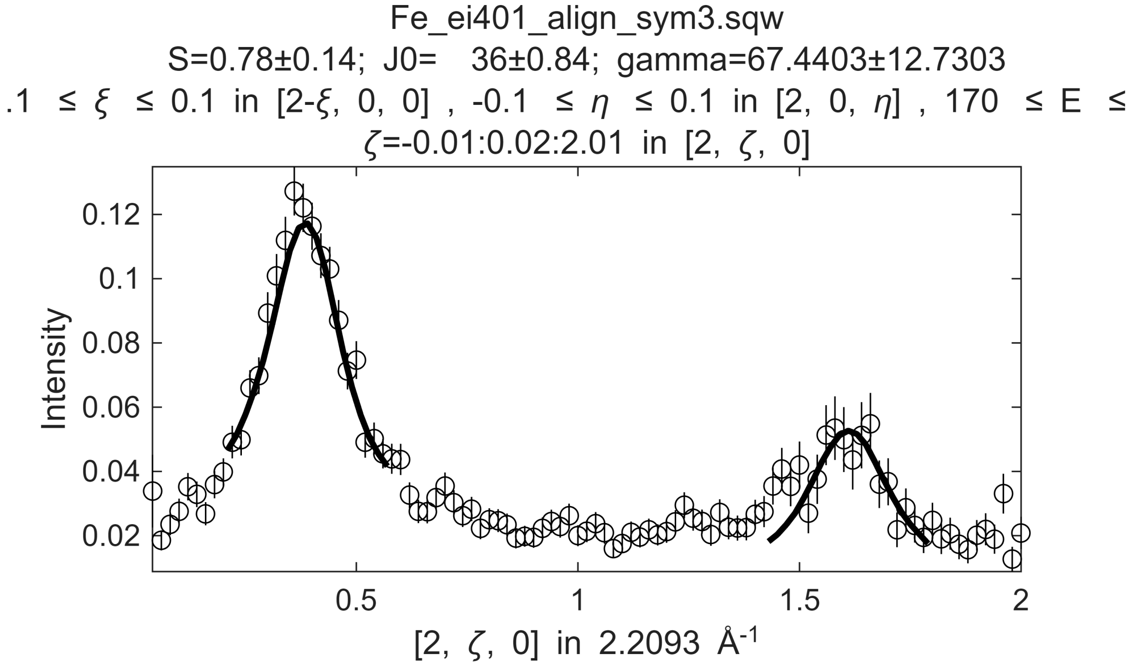

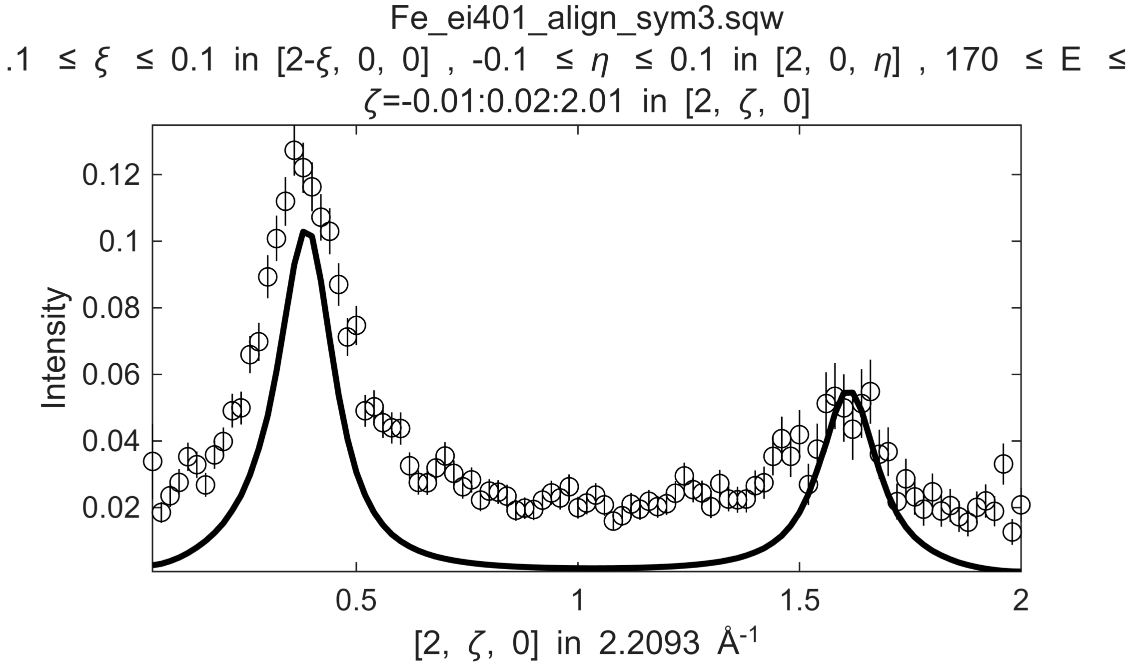


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*** step: 15#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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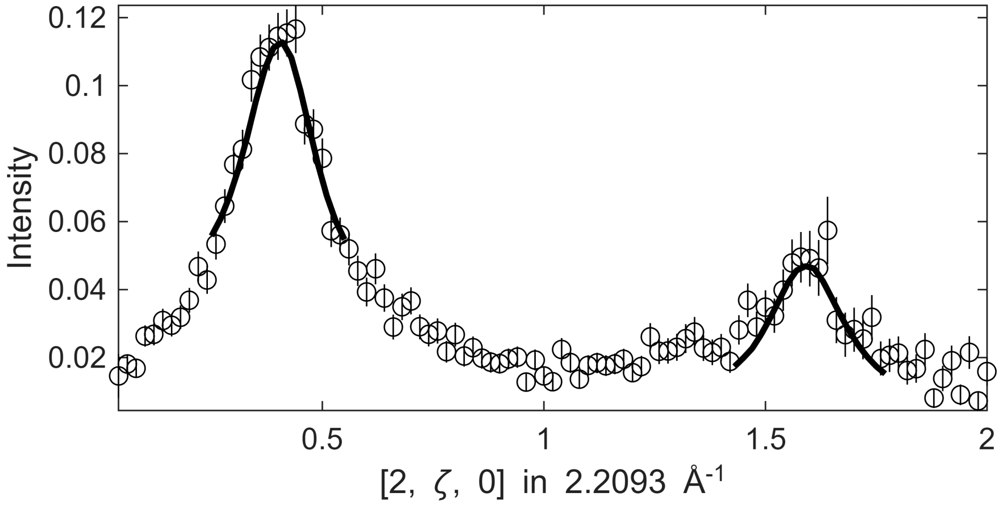
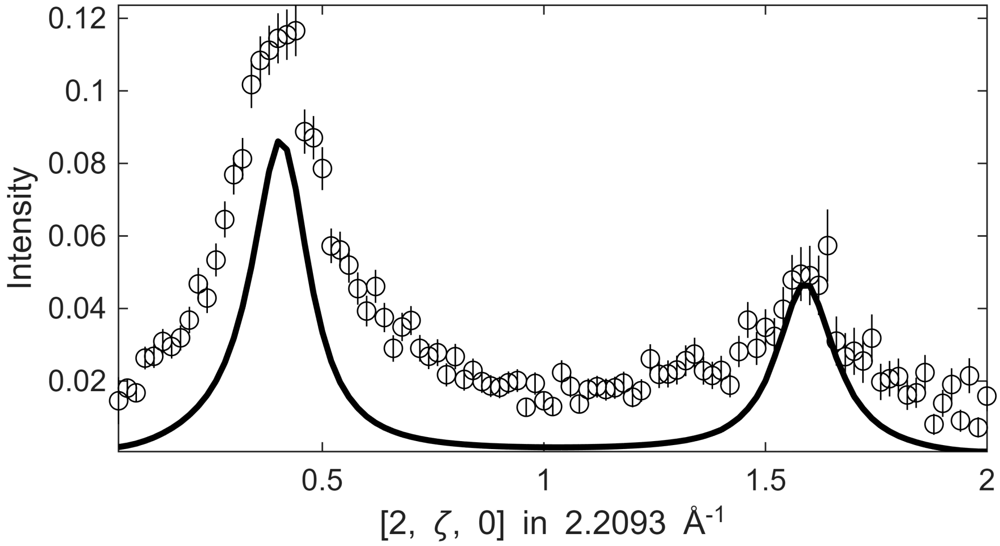
Warning: Dataset:10 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.6502
Warning: Dataset 2:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.92593
```

Fe_ei401_align_sym3.sqw
S=0.65±0.056; J0= 35±0.62; gamma=66.3859±7.32858
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 180 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 180 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

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*** step: 16#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:12 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.9802
Warning: Dataset 2:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.9231
```
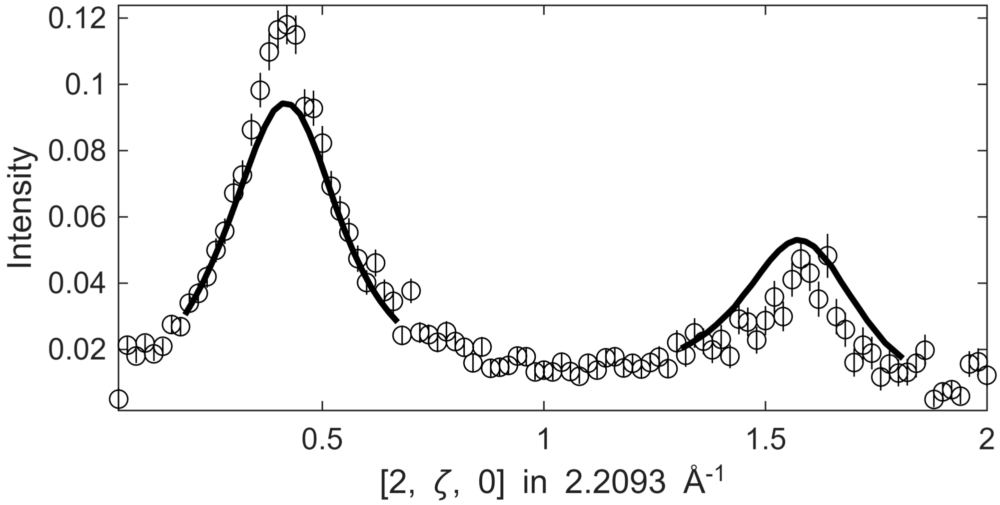
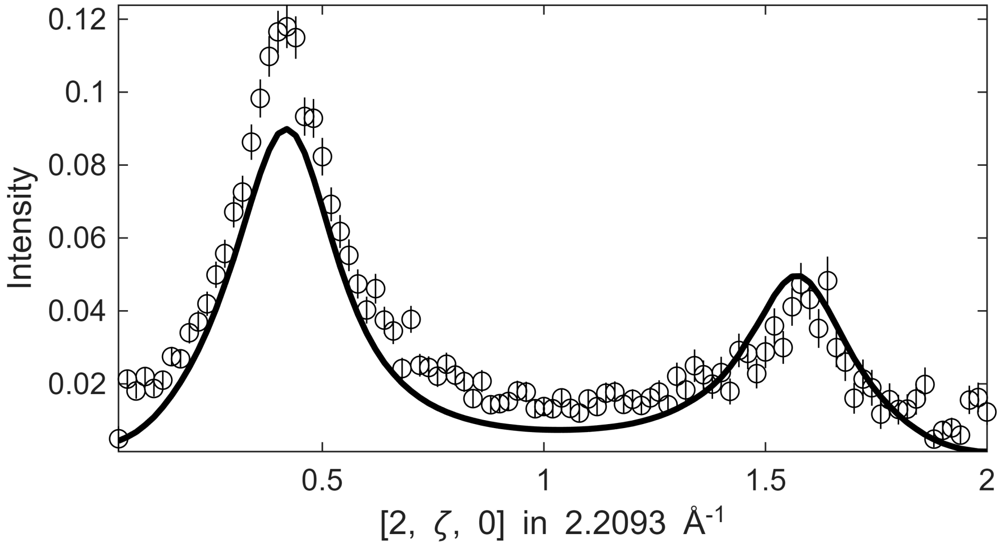

Fe_ei401_align_sym3.sqw
S= 1.7±0.31; J0= 42± 2.7; gamma=192.969±39.6063
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 190 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 190 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

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*** step: 17#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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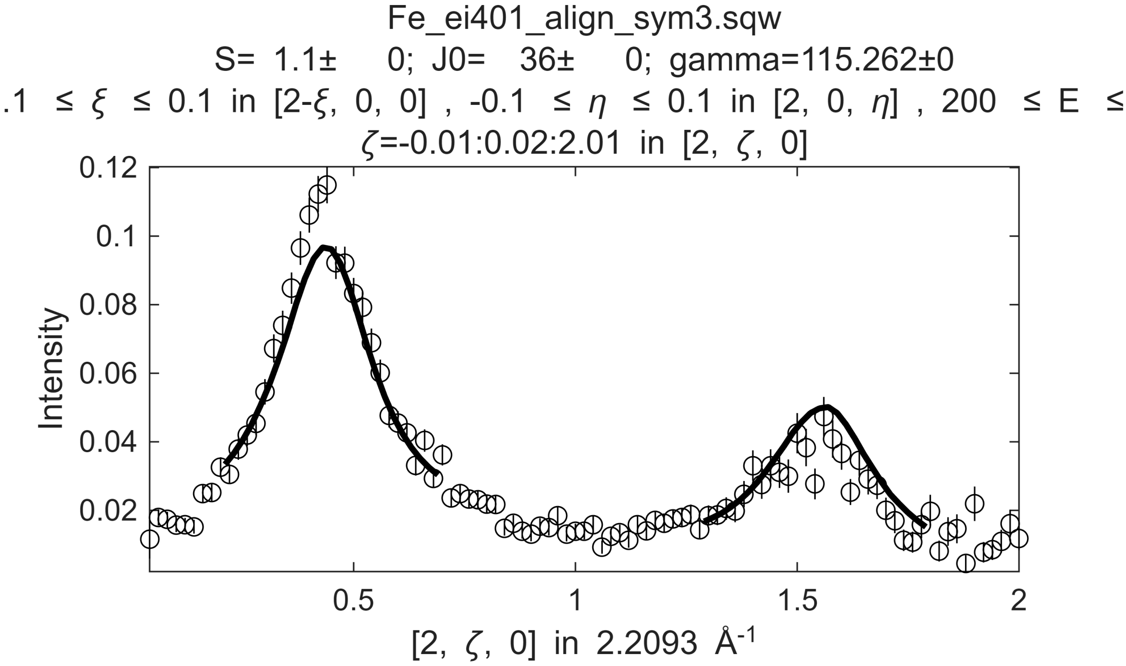
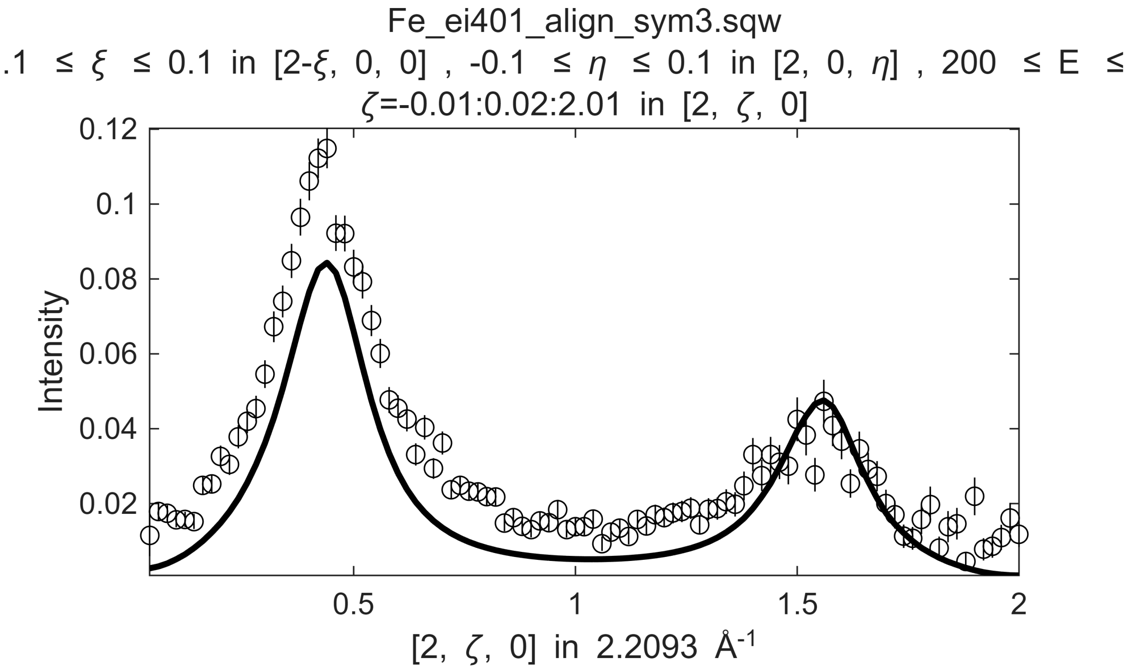
Warning: Dataset:17 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.8053
Warning: Dataset 2:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.64103
Warning: WARNING: Convergence not achieved
```



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*** step: 18#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:18 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 1:2 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.3333
Warning: WARNING: Convergence not achieved
```

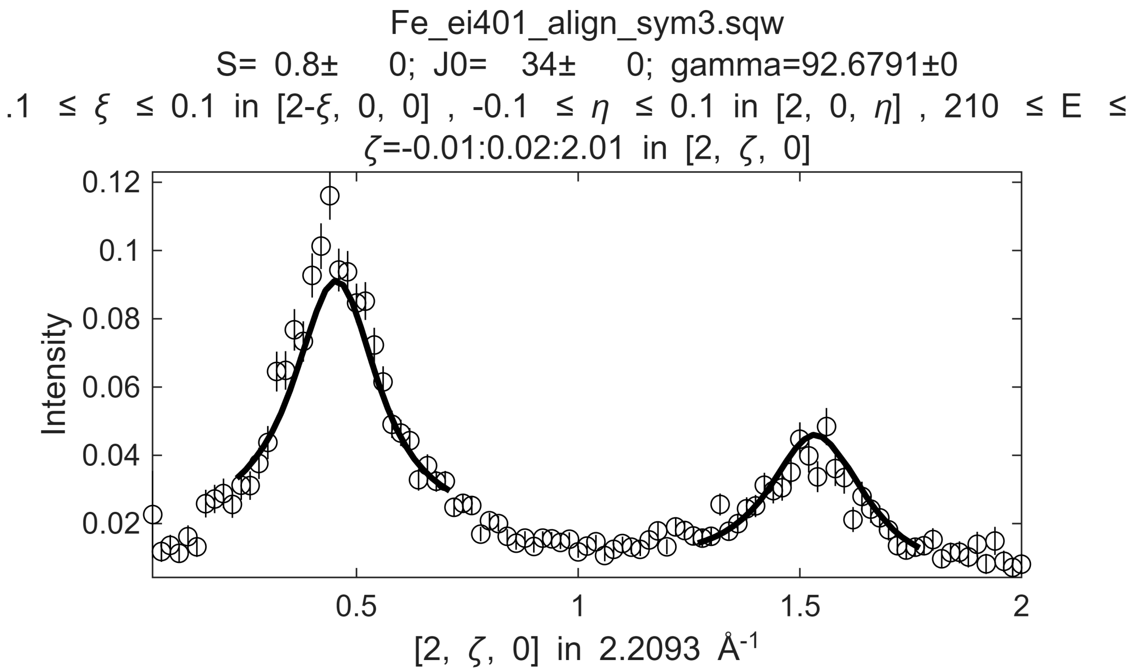

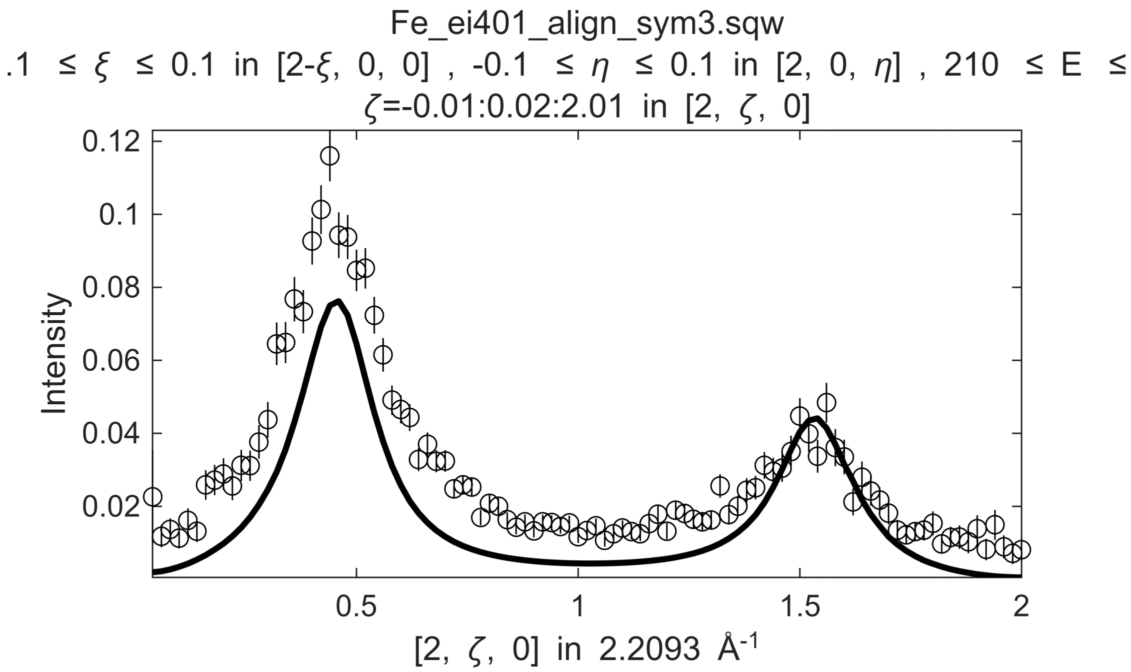


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*** step: 19#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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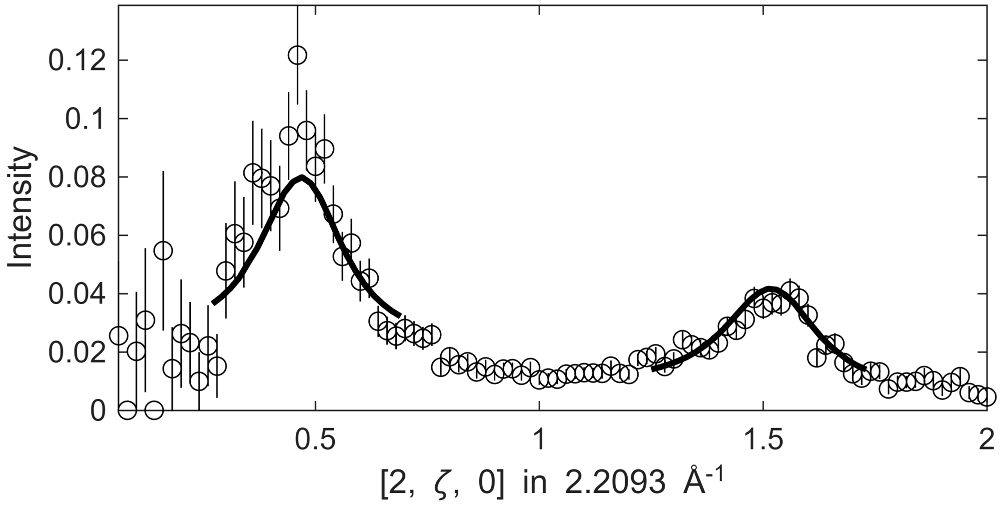
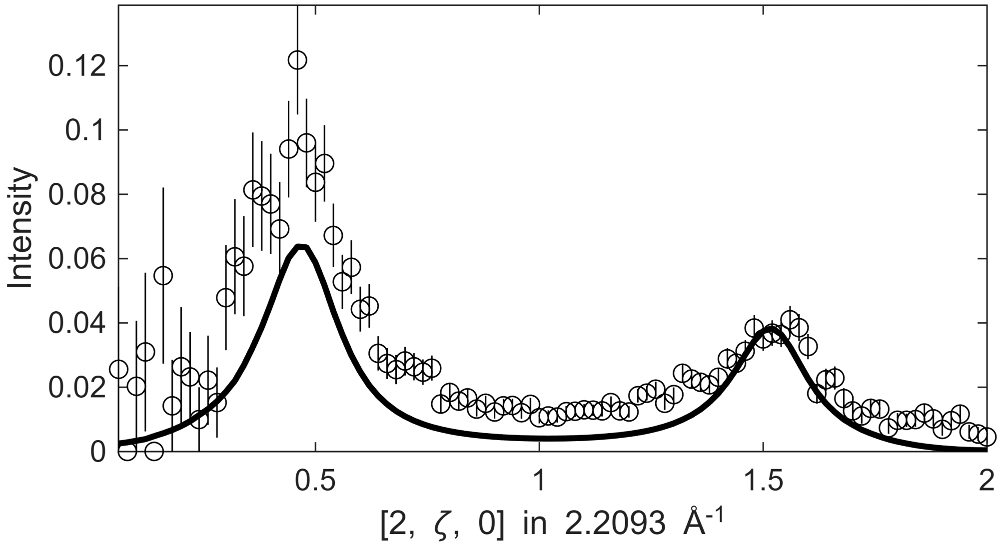
Warning: Dataset:11 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.8152
Warning: Dataset 1:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.75758
```

Fe_ei401_align_sym3.sqw
S=0.66±0.11; J0= 33±0.98; gamma=88.9559±15.5732
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 220 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 220 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

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****************************************************************
*** step: 20#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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Warning: Dataset:11 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.8152
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.64103
```
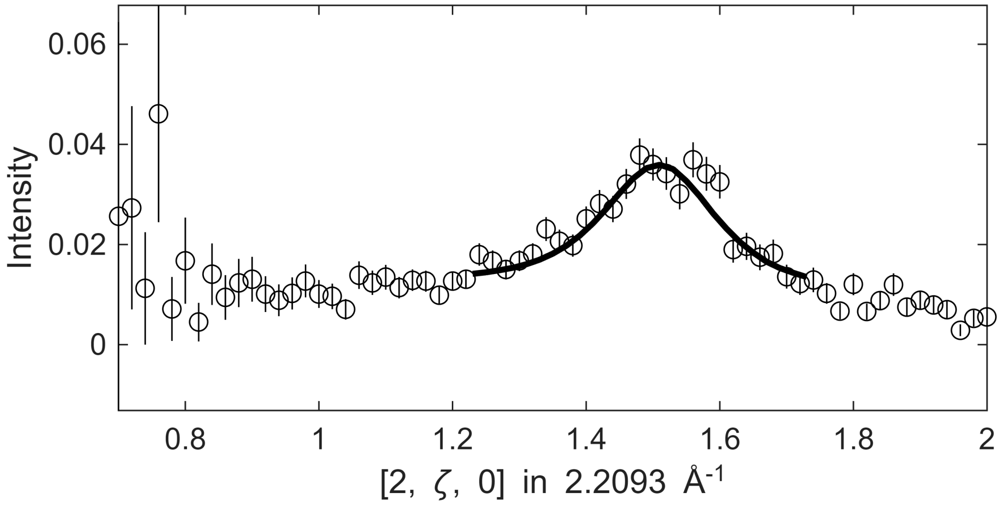
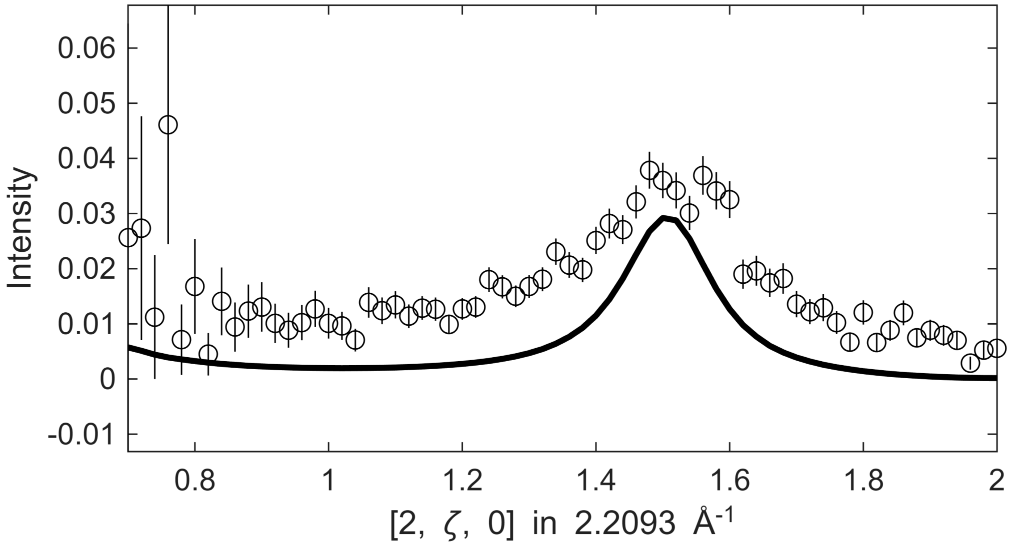

Fe_ei401_align_sym3.sqw
S=0.38±0.059; J0= 32±0.67; gamma=66.1873±11.7939
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 230 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 230 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

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*** step: 21#21  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
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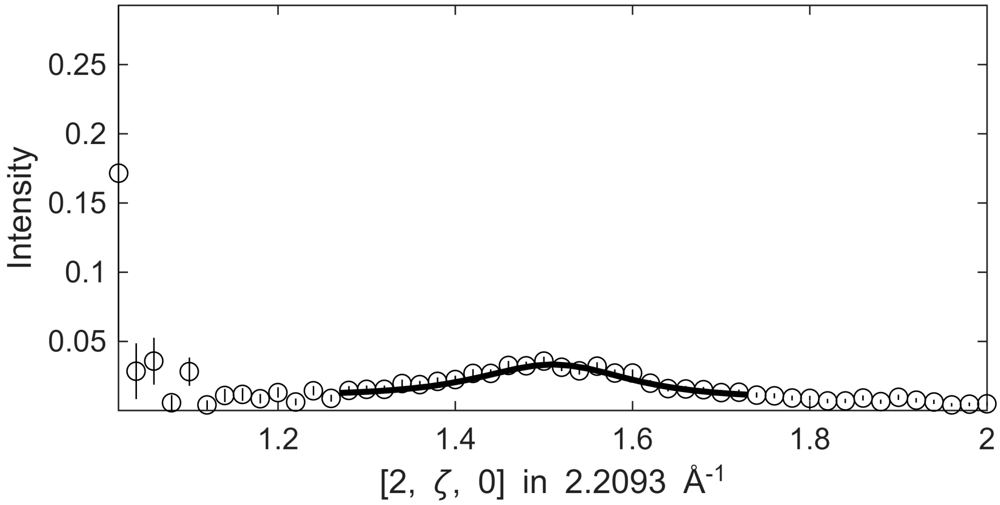
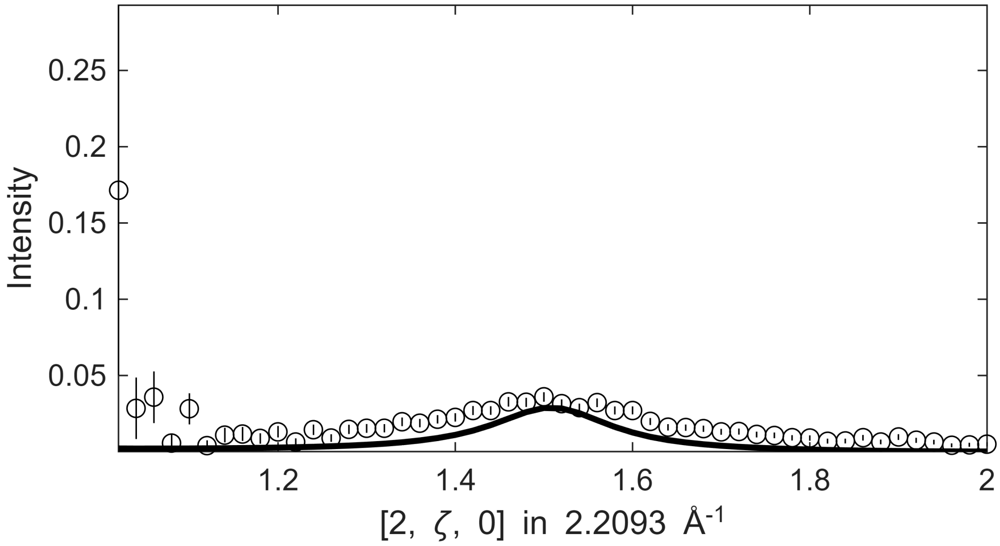
Warning: Dataset:11 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.8152
Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 2.0833
```

Fe_ei401_align_sym3.sqw
S= 0.4±0.087; J0= 33±0.83; gamma=70.8388±16.8897
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 240 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 240 ≤ E ≤
ζ=-0.01:0.02:2.01 in [2, ζ, 0]

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```
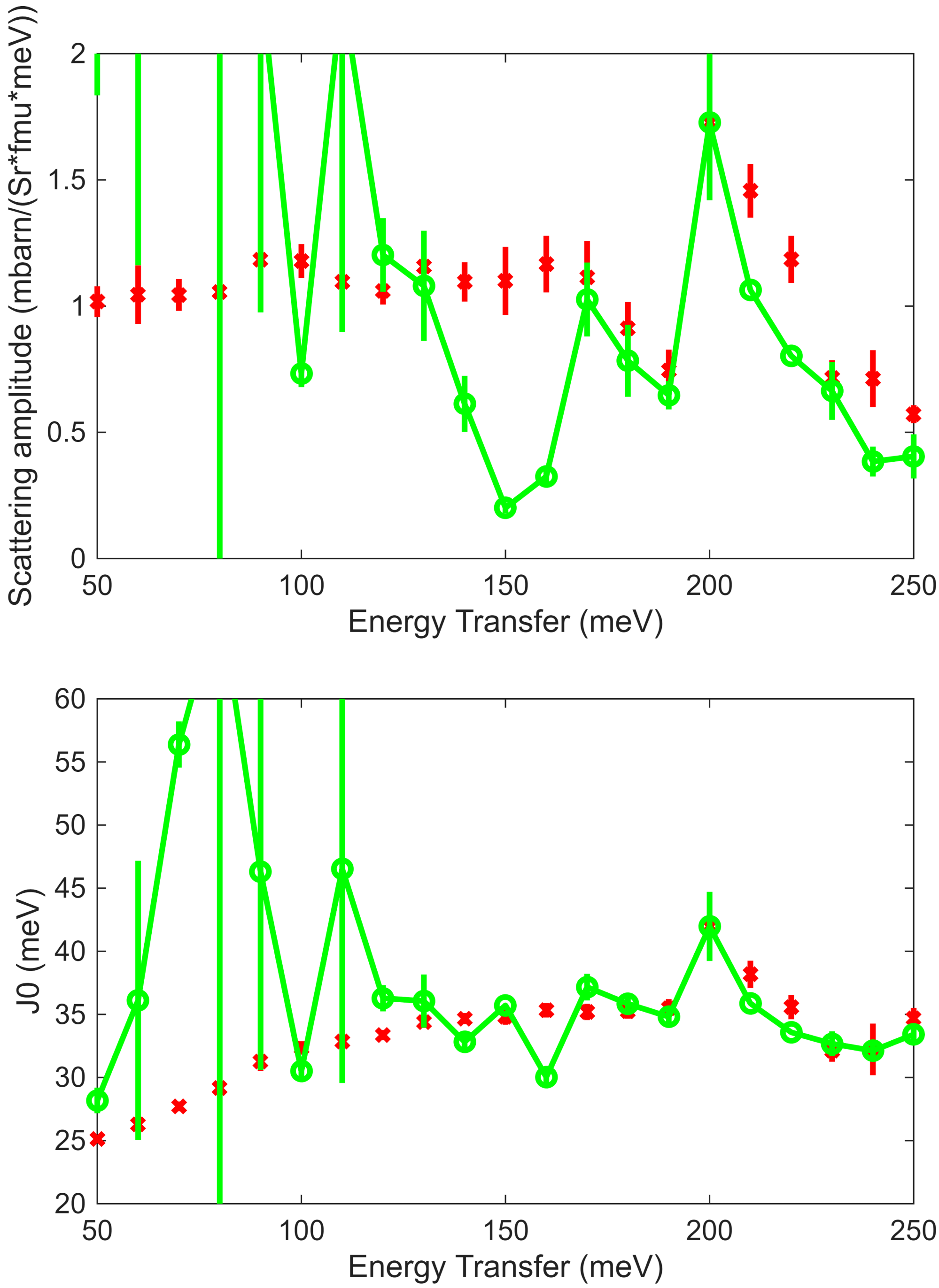
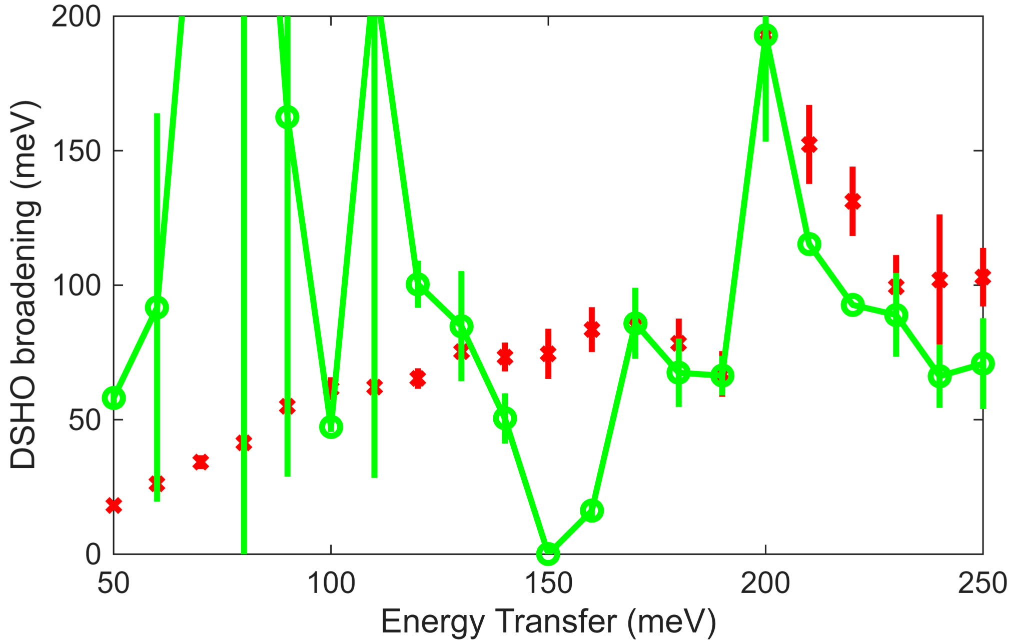
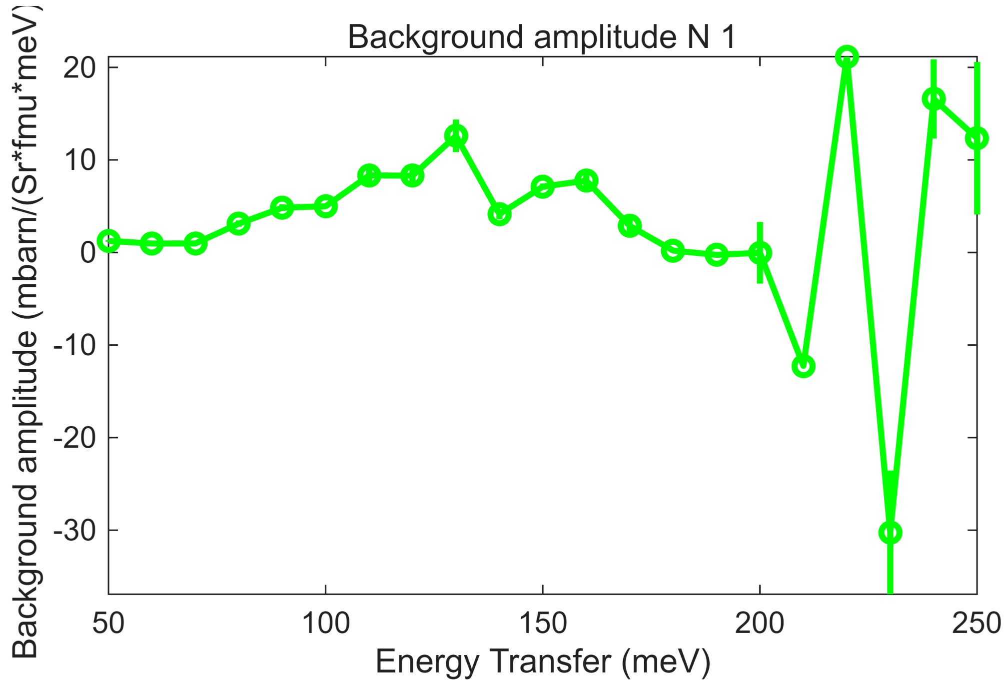

```
plot_SJG_results(fit_res,'g',{'S','J0','gamma'},{[0,2],[20,60],[0,200]},
{Sg,J0,G});
```



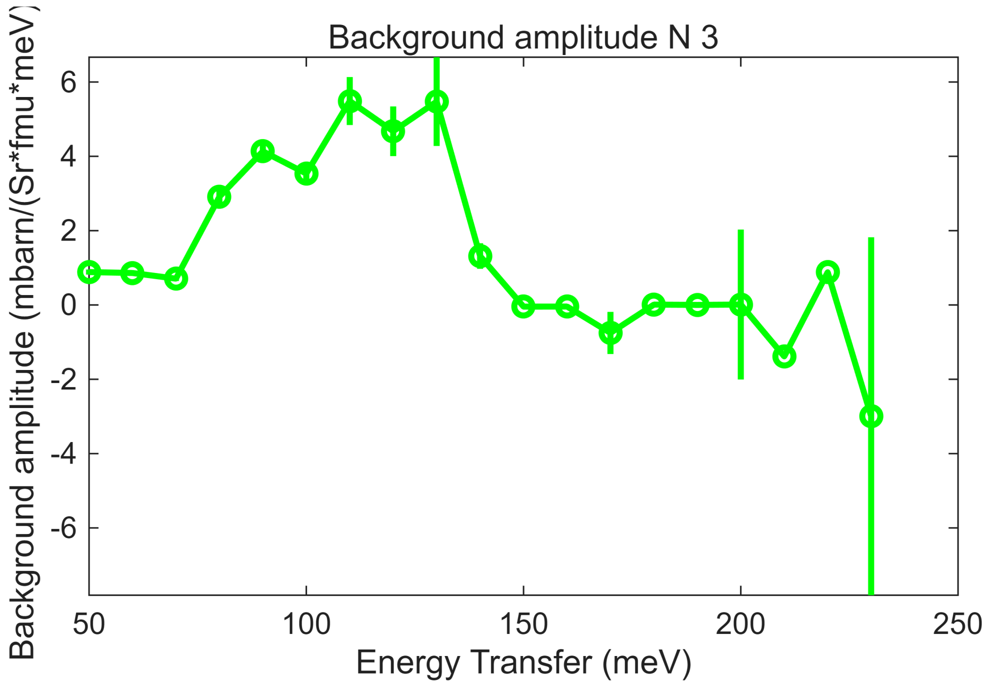
Scattering amplitude (mbarn/(Sr*fmu*meV))
2
1.5
1
0.5
0
50
100
150
200
250
Energy Transfer (meV)
J0 (meV)
60
55
50
45
40
35
30
25
20
50
100
150
200
250
Energy Transfer (meV)




```
bg_data = extract_bg_par(fit_res.peak_fit_par);
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
igure
```




```
pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
igure;
```




```
%fit_res.Speak_varsl = Sg;
%fit_res.J0peak_varsl = J0;
%fit_res.gamma_peak_varsl = G;
fit_res.Speak = Sg;
fit_res.J0peak = J0;
fit_res.gamma_peak = G;
fit_res.peak_fit_FixSlope_par = fit_res.peak_fit_par;
fit_resEi400 = fit_res;
save(fullfile(fit_data_path,fit_data_file),"fit_resEi400");
```